Are Machine Learning Improvements Statistically Real? Dependence-Aware Inference for Reliable Model Comparison on Spatiotemporal Data

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Abstract

Claims that one machine learning model outperforms another usually rest on cross-validated scores and a test that assumes independent observations. On spatial and temporal data this assumption fails twice: random cross-validation leaks information between neighbouring observations, and the errors of the compared models are themselves dependent. We study the second problem for learning procedures that are refitted on each fold. We cast model comparison as inference on the mean of a dependent field of loss differentials with correlated training effects, derive the bias of the paired t-test on fold scores and of a combined variance estimator, and propose DA-P, a test computed from one blocked cross-validation run that adds to a variogram-based test-sampling variance a training-variance term and a Satterthwaite reference distribution, with a three-way decision rule that flags conclusions depending on the unverifiable correlation of training effects. In simulations with exact truth, conventional procedures reject a true null in up to **80%** of replicates. DA-P remains at or below the nominal level with fixed error fields and weakly correlated training effects, including under several misspecifications, and is the best procedure with refitted XGBoost learners, where a spatial prediction comparison test fails; it still rejects about **10%** of true nulls in that setting under strong dependence, a limit we trace to training effects shared by all folds. On weather, pedestrian and yield-monitor data, six of nine conclusions depend on the validation design or the test. Python code is provided.

Keywords: Model comparison, Cross-validation, Spatio-temporal dependence, HAC variance estimation, Variogram, Statistical significance, Trustworthy machine learning

1 Introduction

Empirical machine learning advances one comparison at a time. A new gradient-boosting configuration is reported to beat a random forest, a deep architecture to beat gradient boosting, a feature set to beat another. Such claims rest on a small number of summary statistics (an average accuracy, an F1-score, a root mean squared error) computed by cross-validation, and, in the better cases, on a statistical test or a confidence interval for the difference. The test is meant to answer a simple question: *is the observed improvement larger than what the randomness of the data and of the validation procedure could produce on its own?*

The standard tools for that question [1–4] were developed for observations that are independent and identically distributed (i.i.d.). Much of the data on which machine learning is now used in practice is not. Meteorological, environmental, agricultural, epidemiological and mobility data are collected at locations and times that are close to one another, and nearby observations share unmeasured drivers. Two consequences are well documented. First, random K-fold cross-validation places near-duplicates of test observations in the training set, so that it measures interpolation rather than prediction at new places or times, and is optimistic when the latter is the target [5–8]. Second, the *errors* of a fitted model are themselves dependent, so that n test observations carry much less information than n independent ones; this second problem has been addressed for fixed forecasts and spatial predictions [9, 10], but rarely for learning procedures evaluated by cross-validation.

The first problem has a well-known remedy: blocked (spatial, temporal or spatio-temporal) cross-validation [5, 11, 12]. The second problem is the subject of this paper. Blocking changes *what* is estimated; it does not tell us *how precisely* it is estimated. A practitioner who computes a blocked cross-validation score for two models and then applies a paired t-test, a bootstrap that resamples observations, or a test on fold-wise scores still relies on independence assumptions that the data violate. Whether an apparent improvement is "statistically real" then depends as much on these hidden assumptions as on the models.

Our starting point is the per-observation out-of-fold *loss differential* $d_i = \ell(y_i, \hat{f}_A(x_i)) - \ell(y_i, \hat{f}_B(x_i))$ between two models A and B. The mean of the d_i estimates the expected gain of A over B, and the comparison is a problem of inference on the mean of a dependent random field. Two sources of dependence must be handled at once. (i) Test-point dependence: the d_i inherit the spatial and temporal autocorrelation of the residuals of both models. (ii) Training dependence: all observations of a fold are predicted by the same fitted models, and the models of different folds are fitted on overlapping training sets. Existing procedures handle one source at most. Heteroskedasticity- and autocorrelation-consistent (HAC) variance estimators, as in the Diebold–Mariano test for forecasts [9], and the parametric spatial prediction comparison test of Hering and Genton [10] handle (i) for fixed predictors. The corrected resampled t-test of Nadeau and Bengio [2] handles (ii) for i.i.d. data.

Contributions.

The paper makes five contributions.

1. We formalise model comparison under spatio-temporal dependence as inference on the mean of a loss-differential field with correlated fold-level training effects. We show (Proposition 1) that the paired t-test on fold scores underestimates the variance by exactly the average covariance between fold means, and we derive (Proposition 2) the exact bias of a variance estimator combining test-sampling and training components, which identifies the conditions under which it is unbiased and the direction of its error otherwise.
2. We propose *DA-P*, a dependence-aware test for two models evaluated by blocked cross-validation. It combines (a) a test-sampling variance computed from a separable correlation model fitted to the variograms of the fold-centred loss differentials; (b) a method-of-moments estimate of the training-induced fold variance, with an explicit assumption on the correlation of training effects; and (c) a Welch–Satterthwaite reference distribution. All steps use a single cross-validation run.
3. We attach to the test a three-way decision rule that declares a comparison *indeterminate* when its significance depends on the unverifiable correlation of training effects, and we evaluate it, together with a range-based diagnostic, by simulation.
4. We assess conventional and dependence-aware procedures (paired t, i.i.d. bootstrap, corrected resampled t, block bootstrap, linear mixed model, Diebold–Mariano–HAC, a space–time version of the Hering–Genton test, a kernel variant DA-K, and DA-P) in simulation studies with exact control of the truth, including sensitivity to the assumed training correlation, robustness to anisotropy, covariance shape, heteroscedasticity, non-stationarity and irregular sampling, and gradient-boosting learners whose expected performances are equal by an exchangeability argument.
5. We apply all procedures to three public data sets (hourly weather in New York, hourly pedestrian counts in Melbourne and yield-monitor maize data in Argentina) under random, blocked and forward-chaining validation, with a correction for multiple comparisons and a "seed-null" diagnostic, and we derive practical recommendations. An open-source Python implementation reproduces all results.

The emphasis of the paper is thus not on whether spatial or temporal validation is preferable, a question already addressed in the literature, but on the *validity of the inference* drawn from a given validation design. In the language of the call for this special issue, it is a contribution to measuring the trustworthiness of comparisons between machine learning models in real applications.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 sets up the problem and analyses conventional tests. Section 4 presents DA-P and its decision rule. Sections 5 and 6 report the simulation studies, Section 7 the case studies, and Section 8 discusses recommendations and limitations.

2 Related work

Statistical comparison of learning algorithms.

Dietterich [1] showed that the paired t-test on resampled hold-out scores has an inflated type I error rate, because training sets overlap, and proposed the 5×2 cross-validation test. Nadeau and Bengio [2] derived the corrected resampled t-test, which inflates the variance of repeated train/test differences by the factor $1/J + n_{\text{test}}/n_{\text{train}}$. Bengio and

Grandvalet [13] proved that no unbiased estimator of the variance of K-fold cross-validation exists, which is why every practical procedure relies on a working model or a heuristic. Bouckaert and Frank [14] studied the replicability of these tests, Demšar [3] and Benavoli et al. [4] treated comparisons over multiple data sets, frequentist and Bayesian. More recently, Bayle et al. [15] established central limit theorems for K-fold cross-validation under algorithmic stability, and Bates et al. [16] clarified that cross-validation estimates the average error of the learning *procedure* rather than that of the fitted model, and proposed nested cross-validation intervals. Bouthillier et al. [17] quantified the sources of variance (data split, initialisation, hyperparameter search) in machine learning benchmarks, and Varoquaux [18] showed how wide cross-validation error bars are in small samples. All these works assume independent observations.

Validation under spatial and temporal dependence.

In ecology, remote sensing and the geosciences, a large literature shows that random cross-validation is optimistic for spatially structured data and recommends spatial blocking, buffering or nearest-neighbour distance matching [5–7, 11, 12, 19, 20]. Wadoux et al. [21] argued that, when the goal is the accuracy of a map over a fixed region, design-based probability sampling is preferable to any form of spatial cross-validation. In time series, Bergmeir and Benítez [8] compared blocked and out-of-sample schemes and Bergmeir et al. [22] showed when standard cross-validation remains valid for autoregressive models. These works are concerned with the *point estimate* of predictive performance and its bias; the uncertainty of the comparison between two models, which is the object of this paper, is rarely addressed.

Comparing predictions of dependent data.

Inference on the difference in accuracy between two predictions of dependent data is not new. The Diebold–Mariano (DM) test [9], with the small-sample correction of Harvey et al. [23], applies a HAC (Newey–West [24]) variance estimator to the loss-differential series of two forecasts. Hering and Genton [10] proposed the spatial prediction comparison test, which models the covariance of a spatially correlated loss-differential field with a parametric variogram, and Gilleland et al. [25] showed that the size of such tests is sensitive to serial dependence and to the correlation between competing forecasts. Watson et al. [26] studied prediction and model evaluation for space–time data, with emphasis on how the validation design must match the prediction task. HAC estimation under cross-sectional dependence goes back to Conley [27] and, for panels, to Driscoll and Kraay [28]; the poor size of HAC tests in small effective samples motivated the fixed-b theory of Kiefer and Vogelsang [29]. Block and stationary bootstraps [30–32] provide resampling alternatives.

Positioning of this work.

In view of this literature, our contribution is not the idea of accounting for dependence in the loss differential, which the DM and Hering–Genton tests already do for *fixed* predictors. It lies in four more specific elements. (i) We compare *learning procedures* that are refitted on each training fold of a blocked cross-validation, which adds a training-induced, fold-level variance component, correlated across folds, that is

absent from forecast or map comparisons; we make its estimation and the conditions for its validity explicit (Propositions 1 and 2). (ii) DA-P combines this component with the test-sampling dependence in a single variance estimate computed from one cross-validation run. (iii) The dependence of the test-sampling fluctuations is modelled on the variograms of the *fold-centred* loss differentials, so that it is not confounded with the training effects, and its adequacy is examined under several misspecifications. (iv) The procedure returns diagnostics and a three-way decision that declares a comparison indeterminate when the data contain too few effectively independent regions; the threshold of this rule is calibrated and validated by simulation. We include a space–time version of the Hering–Genton test among the competing procedures in all our experiments.

Trustworthy machine learning and this special issue.

Recent contributions in this journal have advocated statistical tools for the robustness of machine learning models [33], practical safeguards against overfitting [34], and effect-size reporting beyond p-values in large data sets [35]. The present paper complements them by addressing the validity of the comparison step itself when observations are dependent.

3 Problem formulation

3.1 Setting and estimand

We observe n labelled examples $z_i = (x_i, y_i)$, $i = 1, \dots, n$, each attached to a spatial location $s_i \in \mathbb{R}^p$ (typically $p = 2$) and a time $t_i \in \mathbb{R}$; either component may be absent. Two learning procedures A and B (for instance, XGBoost and a random forest, or one learner with two feature sets) are to be compared with a loss ℓ, such as the squared error or the logarithmic loss [36].

The observations are grouped into G spatio-temporal blocks (for instance, square tiles crossed with time windows) and the blocks are allocated to K folds $F_1, \dots, F_K$, with $n_k = |F_k|$. For each fold k, both learners are fitted on a training set $T_k \subseteq \{i \notin F_k\}$, from which observations within a buffer distance of the fold may be removed, and are used to predict the observations of F_k. With $k(i)$ the fold of observation i, the out-of-fold *loss differential* is

$$d_i = \ell\big(y_i, \hat{f}_A^{(k(i))}(x_i)\big) - \ell\big(y_i, \hat{f}_B^{(k(i))}(x_i)\big), \tag{1}$$

so that negative values favour A, and the cross-validated estimate of the gain is $\hat{\Delta} = \bar{d} = n^{-1} \sum_i d_i$.

Definition 1 (Estimand) The *expected cross-validated loss differential* is $\Delta_n = \mathbb{E}[\hat{\Delta}]$, where the expectation is over the data-generating process and the randomness of the learners. It is the difference in expected loss between the two learning procedures, trained on about $n(1 - 1/K)$ observations, when predicting observations that are separated from the training data as prescribed by the blocking design.

The estimand depends on the validation design, as stressed by Bates et al. [16] for i.i.d. data and by the spatial validation literature [5, 21]: with random folds it describes interpolation, with spatial blocks it describes extrapolation to new areas. We take the design as given, chosen to match the intended use of the model, and focus on testing $H_0 : \Delta_n = 0$ against $H_1 : \Delta_n \neq 0$ and on confidence intervals for Δ_n.

3.2 A working model for loss differentials

The difficulty is that the d_i are dependent for two distinct reasons. We make them explicit with the following decomposition.

Definition 2 (Working model)

$$d_i = \Delta_n + \eta_{k(i)} + u_i, \tag{2}$$

where $(\eta_1, \ldots, \eta_K)$ are exchangeable fold effects with $\mathrm{Var}(\eta_k) = \tau^2$ and $\mathrm{Corr}(\eta_k, \eta_l) = \rho$ for $k \neq l$, independent of u, and $u = (u_i)$ is a zero-mean, second-order stationary field with $\mathrm{Cov}(u_i, u_j) = C(s_i - s_j, t_i - t_j)$.

The fold effect η_k captures the variability induced by fitting the two learners on the training set T_k: an unlucky training sample, a different random seed or a different tree structure shifts the losses of all observations of the fold in the same direction. Because the training sets of different folds share a fraction $(K-2)/(K-1)$ of their observations, the fold effects are positively correlated. The field u captures test-sampling variability; it inherits the spatial and temporal autocorrelation of the residuals of both models, which is substantial whenever both models miss a spatially structured signal. Under (2),

$$\mathrm{Var}(\hat{\Delta}) = \underbrace{\frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} C(s_i - s_j, t_i - t_j)}_{V_u} + \underbrace{\tau^2\Big[\sum_k \frac{n_k^2}{n^2} + \rho \sum_{k \neq l} \frac{n_k n_l}{n^2}\Big]}_{V_\eta}. \tag{3}$$

With equal fold sizes, $V_\eta = \tau^2[1/K + \rho(1 - 1/K)]$. Writing $V_u = \sigma_u^2 / n_{\mathrm{eff}}$ defines the *effective sample size* $n_{\mathrm{eff}} = n/[1 + (n-1)\bar{\rho}_u]$, where $\bar{\rho}_u$ is the average correlation between distinct u_i. Under moderate dependence n_{eff} can be orders of magnitude smaller than n.

3.3 Why conventional procedures fail

Procedures that assume independent observations.

The i.i.d. bootstrap of the d_i, the z-test on the d_i and McNemar-type tests estimate σ^2/n and therefore ignore both terms of (3) beyond the diagonal. Their variance is too small by the design-effect factor n/n_{eff}, and by V_η even without spatial dependence.

The paired t-test on fold scores.

The most common procedure computes the K fold means $m_k = n_k^{-1} \sum_{i \in F_k} d_i$ and refers $\bar{m}/(s_m/\sqrt{K})$ to a t_{K-1} distribution, with s_m^2 the sample variance of the m_k. Its weakness is captured by the following identity.

Proposition 1 *Let $m_1, \dots, m_K$ have common mean, average variance $\bar{v} = K^{-1} \sum_k \operatorname{Var}(m_k)$ and average pairwise covariance $\bar{c} = [K(K-1)]^{-1} \sum_{k \neq l} \operatorname{Cov}(m_k, m_l)$. Then*

$$\mathbb{E}\Big[\frac{s_m^2}{K}\Big] - \operatorname{Var}(\bar{m}) = -\bar{c}.$$

Proof $\operatorname{Var}(\bar{m}) = K^{-2}\big[\sum_k \operatorname{Var}(m_k) + \sum_{k \neq l} \operatorname{Cov}(m_k, m_l)\big] = [\bar{v} + (K-1)\bar{c}]/K$. Since the means are equal, $\mathbb{E} \sum_k (m_k - \bar{m})^2 = \sum_k \operatorname{Var}(m_k) - K \operatorname{Var}(\bar{m}) = (K-1)(\bar{v} - \bar{c})$, hence $\mathbb{E}[s_m^2]/K = (\bar{v} - \bar{c})/K$. Subtracting gives the result. □

The paired t-test is therefore valid only if fold means are uncorrelated. Under random K-fold cross-validation of dependent data, every fold is spread over the whole domain, the folds share the same spatial and temporal fluctuations, and $\bar{c}$ is close to $\operatorname{Var}(\bar{m})$ itself: the test then estimates only the small idiosyncratic part of the variance. Overlapping training sets add $\rho\tau^2 > 0$ to $\bar{c}$, which is the mechanism identified by Dietterich [1] and corrected by Nadeau and Bengio [2] for i.i.d. data. Blocking reduces the spatial part of $\bar{c}$, but adjacent blocks of different folds remain correlated, and with $K = 5$ the test has only four degrees of freedom.

Cluster methods.

The block bootstrap and the random-intercept linear mixed model (LMM) treat the G blocks as independent clusters. They capture all within-block covariance, including the fold effects of (2), but ignore covariance between neighbouring blocks, which is non-negligible when the dependence range is comparable with the block size, and they ignore the correlation ρ of fold effects.

Parametric spatial comparison.

The test of Hering and Genton [10], which we extend to space–time with a separable exponential covariance fitted to the spatial and temporal variograms of the centred differentials, estimates V_u from a parametric model. It is designed for fixed predictors and therefore omits V_η, and its validity depends on the adequacy of the covariance model.

Time-only HAC.

Applying the Diebold–Mariano test with a Newey–West variance separately to the series of each site, as is often done for panels of forecasts, handles temporal dependence but ignores the correlation between sites at the same time and the fold effects.

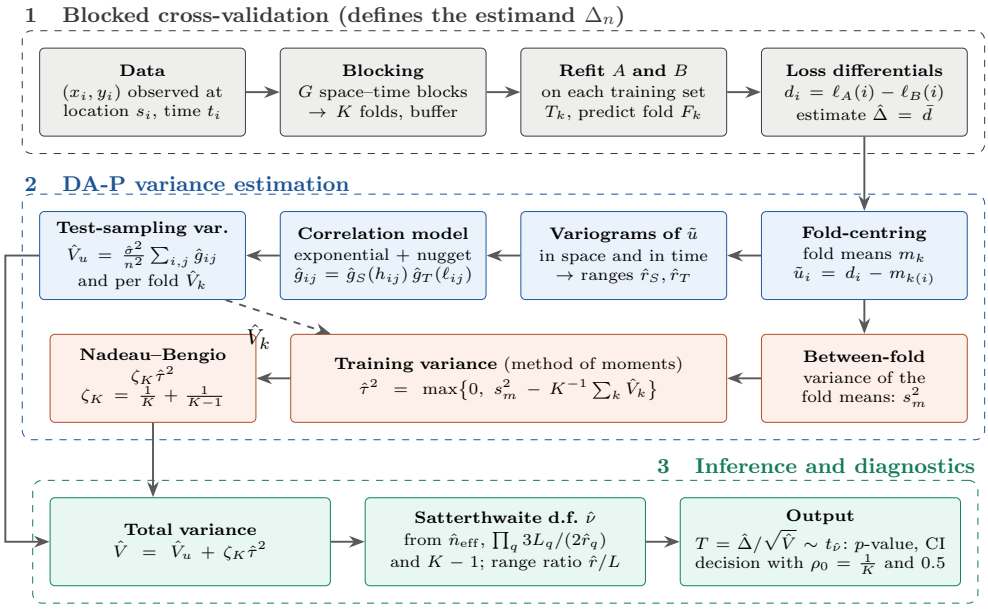


Fig. 1 Architecture of the proposed comparison procedure DA-P. Stage 1 (grey) is a standard blocked cross-validation that yields the per-observation loss differentials d_i and defines the estimand. Stage 2 estimates the test-sampling variance $\hat{V}_u$ (blue branch, read from right to left: fold-centring, variograms, fitted correlation model, variance of the mean) and the training-induced variance $\zeta_K \hat{\tau}^2$ (orange branch). Stage 3 (green) combines them into the total variance, the Satterthwaite degrees of freedom and the test, and returns the diagnostics and the three-way decision of Section 4.7.

4 Dependence-aware inference

The proposed procedure, which we call DA-P (*dependence-aware, parametric*), estimates the two components of (3) from the loss differentials of a single blocked cross-validation run. Figure 1 gives an overview: blocked cross-validation produces the loss differentials (stage 1); two parallel branches estimate the test-sampling variance, from a correlation model fitted to the variograms of the loss differentials, and the training-induced variance, from the dispersion of fold means (stage 2); the two are combined into a variance, a reference distribution, diagnostics and a three-way decision (stage 3). We also describe a kernel variant, DA-K, in which the test-sampling variance is a heteroskedasticity- and autocorrelation-consistent (HAC) estimate; it was our initial proposal and is retained as a point of comparison. The steps are summarised in Algorithm 1 and detailed below.

4.1 Fold-centred differentials

Let m_k be the mean of the d_i in fold k and $\tilde{u}_i = d_i - m_{k(i)}$. Under (2), $\tilde{u}_i = u_i - \bar{u}_{k(i)}$ is free of the fold effects, so that the dependence structure of u can be studied without confusion with the training-induced component.

4.2 Variogram model of the loss differentials

The dependence range of the loss differential is in general different from that of the response: it is the range of the *residual* structure that both models fail to capture. We therefore estimate it from $\tilde{u}$ directly. The directional empirical semivariograms

$$\hat{\gamma}_S(h) = \frac{1}{2|N_S(h)|} \sum_{(i,j)\in N_S(h)} (\tilde{u}_i - \tilde{u}_j)^2, \quad N_S(h) = \{(i,j) : t_i = t_j,\ \|s_i - s_j\| \in I_h\},$$
$$\hat{\gamma}_T(\ell) = \frac{1}{2|N_T(\ell)|} \sum_{(i,j)\in N_T(\ell)} (\tilde{u}_i - \tilde{u}_j)^2, \quad N_T(\ell) = \{(i,j) : s_i = s_j,\ |t_i - t_j| \in I_\ell\},$$

are computed over distance and lag bins I_h, I_ℓ up to a maximum lag (by default half the spatial extent and a quarter of the temporal extent). For purely spatial data, $N_S(h)$ uses all pairs. To each we fit the exponential model with nugget $\gamma(h) = c_0 + c_1\{1 - \exp(-h/a)\}$ by the weighted least squares of Cressie [37], with weights $|N(h)|/\gamma(h)^2$. This yields the correlation functions

$$\hat{g}_S(h) = \frac{\hat{c}_{1,S}}{\hat{c}_{0,S} + \hat{c}_{1,S}} e^{-h/\hat{a}_S} \ (h > 0), \qquad \hat{g}_T(\ell) = \frac{\hat{c}_{1,T}}{\hat{c}_{0,T} + \hat{c}_{1,T}} e^{-\ell/\hat{a}_T} \ (\ell > 0), \tag{4}$$

with $\hat{g}_S(0) = \hat{g}_T(0) = 1$, and the *practical ranges* $\hat{r} = 3\hat{a}$, the distances at which the correlation of the structured component falls to 5% (set to zero when the structured part explains less than 5% of the sill). The *range ratio* $\max_q \hat{r}_q / L_q$, the largest practical range relative to the extent L_q of the domain along the same axis, measures how many effectively independent regions the sample contains.

4.3 Test-sampling variance

DA-P (proposed).

The test-sampling variance and its within-fold analogues are computed from the fitted separable correlation $\hat{g}_{ij} = \hat{g}_S(\|s_i - s_j\|)\, \hat{g}_T(|t_i - t_j|)$:

$$\hat{V}_u = \frac{\hat{\sigma}^2}{n^2} \sum_{i,j} \hat{g}_{ij}, \qquad \hat{V}_k = \frac{\hat{\sigma}^2}{n_k^2} \sum_{i,j\in F_k} \hat{g}_{ij}, \qquad \hat{\sigma}^2 = \frac{1}{n} \sum_i \tilde{u}_i^2. \tag{5}$$

A product of exponential correlations with a nugget is positive definite, so $\hat{V}_u > 0$. Pairs with $\hat{g}_{ij} < 10^{-3}$ are neglected; the others are enumerated with a k-d tree, at a cost proportional to their number. Unlike a kernel estimator, (5) does not truncate the covariance at a bandwidth and does not multiply empirical cross-products, so it is neither biased downward by kernel down-weighting nor degenerate when the range approaches the fold size; its validity rests instead on the adequacy of the exponential model, which we examine under several misspecifications in Section 5.4. It is close in spirit to the spatial prediction comparison test of Hering and Genton [10], with two differences: the variogram is fitted to fold-centred differentials, and the training-induced component below is added.

DA-K (kernel variant).
With bandwidths $b_S = \hat{r}_S$ and $b_T = \hat{r}_T$, the product-Bartlett kernel

$$w_{ij} = \prod_{q=1}^{p} \Big(1 - \frac{|s_{iq} - s_{jq}|}{b_S}\Big)_+ \Big(1 - \frac{|t_i - t_j|}{b_T}\Big)_+ \tag{6}$$

gives the HAC estimates $\hat{V}_u = n^{-2} \sum_{i,j} w_{ij} \tilde{u}_i \tilde{u}_j$ and $\hat{V}_k = n_k^{-2} \sum_{i,j \in F_k} w_{ij} \tilde{u}_i \tilde{u}_j$. Each factor is a one-dimensional Bartlett kernel, whose Fourier transform is non-negative, so the product is positive semi-definite and $\hat{V}_u \geq 0$ (a Bartlett kernel applied to the Euclidean distance in two or more dimensions does not have this property).

4.4 Training-induced variance

Under (2), the between-fold variance of fold means exceeds their within-fold test-sampling variance by the variance of the fold effects. We estimate it by the method of moments,

$$\hat{\tau}^2 = \max\Big\{0,\ s_m^2 - \frac{1}{K} \sum_{k=1}^{K} \hat{V}_k \Big\}. \tag{7}$$

Since $\mathbb{E}[s_m^2]$ subtracts the covariance between fold effects (Proposition 1), $\hat{\tau}^2$ targets $\tau^2(1-\rho)$ rather than τ^2. The correlation ρ is not identifiable from one cross-validation run [13]. For an assumed value ρ_0, the training component of (3) is $V_\eta = \tau^2[1/K + \rho_0(1 - 1/K)] = \zeta_K(\rho_0)\, \tau^2(1 - \rho_0)$, with

$$\zeta_K(\rho_0) = \frac{1/K + \rho_0(1 - 1/K)}{1 - \rho_0}, \qquad \zeta_K(1/K) = \frac{1}{K} + \frac{1}{K - 1}. \tag{8}$$

Following Nadeau and Bengio [2], our default is the test fraction $\rho_0 = n_{\text{test}}/n = 1/K$, for which ζ_K is exactly their inflation $1/J + n_{\text{test}}/n_{\text{train}}$ with $J = K$ splits. The final variance estimate is

$$\hat{V} = \hat{V}_u + \zeta_K(\rho_0)\, \hat{\tau}^2. \tag{9}$$

4.5 What the variance estimator targets

The following proposition makes explicit the conditions under which (9) is unbiased, and the direction of its bias when they fail. For either estimator of the test-sampling variance, we consider the untruncated estimator $\hat{\tau}^2_{\text{raw}} = s_m^2 - K^{-1} \sum_k \hat{V}_k$ and $\hat{V}_{\text{raw}} = \hat{V}_u + \zeta_K(\rho_0) \hat{\tau}^2_{\text{raw}}$.

Proposition 2 *Under the working model (2) with equal fold sizes, let $\bar{m}_k^u$ be the fold means of u, $v_u = K^{-1} \sum_k \operatorname{Var}(\bar{m}_k^u)$, $c_u = [K(K-1)]^{-1} \sum_{k \neq l} \operatorname{Cov}(\bar{m}_k^u, \bar{m}_l^u)$, and let $\beta_u = \mathbb{E}[\hat{V}_u] - V_u$ and $\bar{\beta} = K^{-1} \sum_k \{\mathbb{E}[\hat{V}_k] - \operatorname{Var}(\bar{m}_k^u)\}$ be the biases of the kernel estimators. Then*

$$\mathbb{E}[\hat{V}_{\text{raw}}] - \operatorname{Var}(\hat{\Delta}) = \underbrace{\beta_u - \zeta_K(\rho_0)\,(c_u + \bar{\beta})}_{\textit{test-sampling part}} + \underbrace{\tau^2 \Big[(1-\rho)\,\zeta_K(\rho_0) - \tfrac{1}{K} - \rho\big(1 - \tfrac{1}{K}\big)\Big]}_{\textit{training part}}. \tag{10}$$

In particular, $\hat{V}_{\mathrm{raw}}$ is unbiased if (i) $\rho_0 = \rho$, (ii) fold means of u are uncorrelated across folds ($c_u = 0$) and (iii) the kernel estimators are unbiased ($\beta_u = \bar{\beta} = 0$). The training part is negative, i.e. the variance is underestimated, if and only if $\rho > \rho_0$.

Proof The fold means are $m_k = \Delta_n + \eta_k + \bar{m}_k^u$ with η independent of u, so their average variance is $\tau^2 + v_u$ and their average covariance $\rho\tau^2 + c_u$. By the identity in the proof of Proposition 1, $\mathbb{E}[s_m^2] = (1-\rho)\tau^2 + v_u - c_u$, and by definition $\mathbb{E}[K^{-1}\sum_k \hat{V}_k] = v_u + \bar{\beta}$, hence $\mathbb{E}[\hat{\tau}^2_{\mathrm{raw}}] = (1-\rho)\tau^2 - c_u - \bar{\beta}$. With equal fold sizes, (3) reads $\mathrm{Var}(\hat{\Delta}) = V_u + \tau^2[1/K + \rho(1-1/K)]$ and $\mathbb{E}[\hat{V}_u] = V_u + \beta_u$. Substituting gives (10). Writing $g(r) = 1/K + r(1-1/K)$, the training part is $\tau^2[(1-\rho)g(\rho_0)/(1-\rho_0) - g(\rho)]$, which vanishes at $\rho = \rho_0$ and is strictly decreasing in ρ, with derivative $-\tau^2[g(\rho_0)/(1-\rho_0) + 1 - 1/K] < 0$. □

Proposition 2 separates three sources of error, each of which is evaluated empirically in Section 5. *Training correlation.* For $K = 5$ and $\rho_0 = 1/K$, the training part equals $\tau^2(0.25 - 1.25\rho)$: assuming $\rho_0 = 0.2$ when $\rho = 0.5$ leaves $0.375\tau^2$ of the $0.6\tau^2$ training variance unaccounted for, whereas $\rho < \rho_0$ errs on the conservative side. The sensitivity of the test to this choice is reported in Section 5.3. *Cross-fold covariance.* c_u is the covariance between test-sampling fluctuations of different folds; it is reduced by allocating whole blocks to folds and by buffers, and vanishes when blocks are larger than the dependence range. Because it enters with a negative sign, residual cross-fold dependence makes the test anti-conservative. *Estimation bias of the test-sampling variance.* For DA-K, fold-centring removes the fold-level average of u from $\tilde{u}$ and the Bartlett kernel down-weights covariances near the bandwidth; both make β_u and $\bar{\beta}$ negative when the dependence range is not small relative to the blocks and to the domain. For DA-P, the biases come from the variogram fit (in particular from ranges that are too long to be estimated within the domain) and from misspecification of the exponential model. The term $-\zeta_K\bar{\beta}$ partially offsets β_u: the part of the fold-level variance missed by the within-fold estimators is recovered by $\hat{\tau}^2$. Finally, truncation at zero, $\hat{\tau}^2 = \max\{0, \hat{\tau}^2_{\mathrm{raw}}\}$, can only increase the variance, so $\mathbb{E}[\hat{V}] \geq \mathbb{E}[\hat{V}_{\mathrm{raw}}]$; its effect on the reference distribution is discussed next.

4.6 Reference distribution

HAC tests are known to over-reject when the effective sample size is small [29], and $\hat{\tau}^2$ rests on only $K-1$ degrees of freedom. We approximate the distribution of $\hat{V}$ by a scaled χ^2 with Welch–Satterthwaite [38] degrees of freedom

$$\hat{\nu} = \frac{\hat{V}^2}{\hat{V}_u^2/\nu_u + (\zeta_K\hat{\tau}^2)^2/(K-1)}, \qquad \nu_u = \min\Big\{\hat{n}_{\mathrm{eff}} - 1,\ \prod_q \frac{3L_q}{2\hat{r}_q}\Big\}, \tag{11}$$

with $\hat{n}_{\mathrm{eff}} = n^{-1}\sum_i \tilde{u}_i^2/\hat{V}_u$ truncated to $[2, n]$, where the product runs over the spatial and temporal axes with a positive practical range, and L_q is the extent of the domain along axis q. The second term of ν_u is the classical approximation to the degrees of freedom of a kernel spectral estimate [39] with a Bartlett window of width $\hat{r}_q$: for a lag window k of bandwidth b on a record of length L, $\mathrm{Var}(\hat{V})/V^2 \approx 2(b/L)\int k^2$, with

$\int k^2 = 2/3$, and the factors multiply across axes. It is used for both variants as an approximate count of the independent regions available to estimate the dependence, and caps the degrees of freedom when the range is not small compared with the domain. The test statistic $T = \hat{\Delta}/\sqrt{\hat{V}}$ is referred to a Student $t_{\hat{\nu}}$ distribution, and the $(1-\alpha)$ confidence interval is $\hat{\Delta} \pm t_{\hat{\nu},1-\alpha/2}\sqrt{\hat{V}}$.

When $\hat{\tau}^2_{\text{raw}} \leq 0$ the training term vanishes and $\hat{\nu} = \nu_u$, which may be large even though only K folds inform the training component. Truncation occurs when the observed dispersion of fold means is no larger than its predicted test-sampling part, that is, when the data show no evidence of a training effect; the variance itself is then not underestimated on average (Proposition 2), but its sampling variability is understated. Section 5 reports the rejection rate conditionally on truncation to check that this boundary case does not cause over-rejection.

4.7 A three-way decision rule

Proposition 2 identifies the assumed training correlation ρ_0 as the one input of the procedure that cannot be checked from the data, and the one whose misspecification makes the test anti-conservative. We therefore attach to DA-P a decision rule with three outcomes that makes the dependence of the conclusion on ρ_0 explicit:

$$\text{decision} = \begin{cases} \text{reject } H_0 & \text{if } p(\rho_0 = 1/K) < \alpha \text{ and } p(\rho_0 = 0.5) < \alpha, \\ \text{indeterminate} & \text{if } p(\rho_0 = 1/K) < \alpha \leq p(\rho_0 = 0.5), \\ \text{do not reject } H_0 & \text{otherwise.} \end{cases} \tag{12}$$

An "indeterminate" outcome states that the apparent improvement is significant only if the training effects of different folds are assumed to be weakly correlated, and would not survive a correlation of 0.5; it is not counted as evidence of superiority. Because $\zeta_K(\rho_0)$ is increasing in ρ_0, the second p-value is never smaller than the first.

We also report the range ratio $\max_q \hat{r}_q/L_q$ and the effective sample size $\hat{n}_{\text{eff}}$ as diagnostics of the *information* available for the comparison. We initially intended to declare comparisons indeterminate when the range ratio exceeded a threshold, as suggested by the behaviour of DA-K; the simulations of Section 5.5 show that, for DA-P, large range ratios reduce power but do not cause over-rejection, while the residual over-rejection of Experiment 2 occurs at intermediate ratios. A threshold on the range ratio would therefore not protect the validity of DA-P, and we do not use one.

4.8 Properties

Remark 1 (Special cases) If the fitted variograms show no structure ($\hat{c}_1 = 0$ in both directions) and $\hat{\tau}^2 = 0$, then $\hat{V} = n^{-2}\sum_i \tilde{u}_i^2$ and DA-P reduces to the one-sample t-test on the d_i; the same holds for DA-K with zero bandwidths. If the data have no spatial (or no temporal) coordinate, the corresponding factor of $\hat{g}_{ij}$ or of (6) is omitted. Both variance estimates are non-negative.

Remark 2 (Consistency) Under increasing-domain asymptotics, the DA-P variance is consistent when the exponential model is correctly specified and its parameters are consistently

Algorithm 1 DA-P test for $H_0 : \Delta_n = 0$

Require: loss differentials d_i, folds $k(i)$, coordinates (s_i, t_i), level α

1: $m_k \leftarrow$ mean of d_i over F_k; $\tilde{u}_i \leftarrow d_i - m_{k(i)}$; $\hat{\Delta} \leftarrow n^{-1} \sum_i d_i$
2: Fit exponential models to the spatial and temporal empirical variograms of $\tilde{u}$; correlations $\hat{g}_S, \hat{g}_T$ and ranges $\hat{r}_S, \hat{r}_T$
3: Compute $\hat{V}_u$ and $\hat{V}_1, \ldots, \hat{V}_K$ by (5) ▷ DA-K: kernel (6) instead
4: $\hat{\tau}^2 \leftarrow \max\{0, s_m^2 - K^{-1} \sum_k \hat{V}_k\}$; $\hat{V} \leftarrow \hat{V}_u + \zeta_K(\rho_0)\hat{\tau}^2$ ▷ default $\rho_0 = 1/K$
5: Compute $\hat{n}_{\text{eff}}$, ν_u and $\hat{\nu}$ by (11), and the range ratio $\max_q \hat{r}_q / L_q$
6: $p \leftarrow 2 \Pr(t_{\hat{\nu}} > |\hat{\Delta}|/\sqrt{\hat{V}})$; interval $\hat{\Delta} \pm t_{\hat{\nu},1-\alpha/2}\sqrt{\hat{V}}$
7: Repeat lines 4–6 with $\rho_0 = 0.5$
8: **return** p, the interval, the diagnostics $(\hat{r}_S, \hat{r}_T, \hat{n}_{\text{eff}}, \max_q \hat{r}_q / L_q)$ and the decision (12)

estimated [40], and the DA-K variance is a Conley-type HAC estimator [27], consistent under mixing conditions when the bandwidths grow with, but more slowly than, the size of the domain [39]. Our variogram parameters are data-dependent and our variance includes a component, $\hat{\tau}^2$, estimated from only K folds; we therefore do not claim exact asymptotic validity, and instead evaluate the finite-sample behaviour of the whole procedure by simulation with known truth.

Remark 3 (Scope) DA-P requires only the per-observation losses of the two models and their coordinates; it is agnostic to the learners, to the loss and to the blocking scheme. The same construction applies to any additive performance measure (squared error, absolute error, log loss, Brier score, 0–1 loss). Non-additive measures such as the AUC or the F1-score would require a linearisation (influence function) first.

5 Simulation study I: controlled loss-differential fields

The first study isolates the inferential problem from the learning problem. Prediction errors are simulated directly, so that the true expected loss differential is known exactly and the dependence of the d_i is controlled.

5.1 Design

Fields.

Observations lie on a 12×12 grid of sites observed at 24 time points ($n = 3456$). All fields are separable Gaussian fields with unit variance, exponential spatial correlation $\exp(-h/a)$ and AR(1) temporal dependence with coefficient ϕ. Four dependence levels are considered: *none* ($a = 0$, $\phi = 0$), *weak* ($a = 1$, $\phi = 0.3$), *moderate* ($a = 2$, $\phi = 0.6$) and *strong* ($a = 4$, $\phi = 0.8$). The corresponding practical ranges are $3a = 0, 3, 6, 12$ grid units in space and $-3/\log\phi = 0, 2.5, 5.9, 13.4$ time steps; at the strong level the range is as large as the domain.

Losses.

With ε, e_A and e_B three independent fields, the response noise is ε, model A has errors e_A and model B has errors $\sqrt{1+\delta}\, e_B$. Under squared loss,

$$d_i = (\varepsilon_i - e_{A,i})^2 - (\varepsilon_i - \sqrt{1+\delta}\, e_{B,i})^2 + \eta_{k(i)}, \qquad \mathbb{E}[d_i] = -\delta,$$

so that $\Delta_n = -\delta$ exactly. The fold effects $\eta_k = \sigma_\eta(\sqrt{\rho}\, g + \sqrt{1-\rho}\, h_k)$, with $g, h_1, \ldots, h_K$ i.i.d. standard normal, $\sigma_\eta = 0.05$ and $\rho = 0.25$, mimic the training-induced variability of (2), including its correlation across folds. We use $\delta = 0$ (null) and $\delta \in \{0.1, 0.2, 0.4\}$ (alternatives).

Validation and procedures.

The domain is cut into 4×4-site tiles crossed with 6-step time windows ($G = 36$ blocks), which are allocated to $K = 5$ folds with balanced sizes. We compare the paired t-test on fold means, the i.i.d. bootstrap, the block bootstrap and a random-intercept LMM (both with the 36 blocks as clusters), the Diebold–Mariano test with a site-wise Newey–West variance (DM–HAC), the space–time Hering–Genton test, the kernel variant DA-K and the proposed DA-P. Ablations of DA-P remove the training term or replace the Satterthwaite reference by a normal distribution; a kernel estimator with a fixed bandwidth equal to the block size is also included. Bootstraps use 999 resamples. There are 1000 replicates per dependence level under the null and 300 under each alternative, so that the Monte Carlo standard error of a rejection rate of 5% is 0.7 percentage points.

Pilot studies and choice of the variance estimator.

All design choices were made in pilot studies on random seeds disjoint from those of the main studies, whose scripts and outputs are included in the code repository. Our initial proposal was the kernel estimator DA-K, with bandwidths equal to the estimated practical ranges (multipliers between 0.5 and 2 changed rejection rates by at most 2.7 points). The pilot revealed that its residual over-rejection under moderate and strong dependence came almost entirely from replicates in which $\hat{\tau}^2$ was truncated at zero: without the compensation provided by $\hat{\tau}^2$, the downward bias of the fold-centred kernel estimate (Proposition 2) was no longer offset. Replacing the kernel by the fitted correlation model removed this problem in pilot versions of both simulation studies, with maximum pilot rejection rates of 4.0% in Experiment 1 and 9.2% in Experiment 2, against 16.3% and 14.2% for DA-K; DA-P was therefore adopted before the main studies were run.

5.2 Results

Type I error.

Table 1 and Fig. 2 report rejection rates under the null. Procedures that ignore dependence fail quickly: the i.i.d. bootstrap rejects a true null in 19% of replicates under weak dependence and in 80% under strong dependence, and the time-only DM–HAC test, which ignores correlation between sites, behaves almost as badly (17% and 70%).

The two cluster procedures, block bootstrap and LMM, drift upwards as the range of dependence approaches the block size (15% and 14% at the moderate level, 27% and 25% at the strong level), because they treat neighbouring blocks as independent, and they reject 9% and 7% of true nulls even without dependence, as they ignore the correlation of fold effects. The paired t-test on blocked fold means, the best conventional procedure in line with Proposition 1, reaches 11% at the moderate and 18% at the strong level, and the kernel variant DA-K 10% and 13%.

DA-P keeps the rejection rate at or below the nominal level at every dependence level (3.1%, 4.2%, 3.4% and 2.4%); it is slightly conservative, most at the strong level, where the practical range is as long as the domain and the variance is inflated by the small number of independent regions. The space–time Hering–Genton test, whose exponential model is correctly specified in this design, is also valid (4.4%, 5.5%, 4.4%, 2.6%). Removing the training term from DA-P increases its rejection rate by 1–2 points, with a maximum of 5.7%; in this design the training variance is small ($\sigma_\eta = 0.05$), and its role is examined in Section 5.3 and in Experiment 2.

Boundary case $\hat{\tau}^2 = 0$***.***

Table 2 reports the rejection rate of DA-P according to whether the training variance was truncated at zero. Truncation is frequent (63–76% of replicates), because the training effects are small compared with the test-sampling variability of fold means; the rejection rate in truncated replicates (2.5–6.1%) remains close to the nominal level, and that in the other replicates is low. The large degrees of freedom obtained after truncation therefore do not cause over-rejection here; in DA-K, by contrast, truncated replicates accounted for nearly all false positives (see the pilot paragraph above).

Coverage, width and power.

Figure 3 reports the coverage of the 95% intervals separately for each true gap δ and dependence level. DA-P covers the true Δ_n in 95–99% of replicates in every cell, and Hering–Genton in 94–98%, whereas the i.i.d. bootstrap and DM–HAC fall to 17–32% under strong dependence and the cluster procedures to 67–78%. The DA-P intervals are wider than those of the cluster procedures (Fig. 2b), which is the price of validity. Raw rejection rates under the alternatives (Fig. 4) are not comparable across procedures whose sizes differ by an order of magnitude; we therefore also report size-adjusted power, using as critical value the 5% quantile of each procedure's null p-values (Table 3). Under moderate dependence, DA-P, Hering–Genton and the block bootstrap have the same size-adjusted power (46% at $\delta = 0.4$), above that of the paired t-test and DA-K (29%); without dependence, DA-P (56% at $\delta = 0.2$) is less efficient than the cluster procedures (81%) and Hering–Genton (63%), because its training term is estimated from $K - 1 = 4$ degrees of freedom. Its slight conservativeness also lowers its raw power, for example 48% against 63% for Hering–Genton at $\delta = 0.2$ without dependence.

5.3 Sensitivity to the correlation of training effects

Proposition 2 shows that the variance is underestimated when the true correlation ρ of fold effects exceeds the assumed value ρ_0. To quantify this, we increased the

Table 1 Experiment 1: rejection rate (%) of a true null at $\alpha = 0.05$ (1000 replicates per level) and coverage (%) of the 95% confidence interval (all 1900 replicates per level). Bold: rejection rate outside the 95% Monte Carlo band $[3.6, 6.4]$, above (anti-conservative) or below (conservative).

	Type I error (%)				Coverage (%)			
Procedure	None	Weak	Mod.	Strong	None	Weak	Mod.	Strong
Paired t (folds)	**6.6**	**6.7**	**10.7**	**17.9**	93.5	93.6	89.2	81.4
i.i.d. bootstrap	**8.1**	**19.4**	**57.4**	**80.3**	91.8	79.7	43.0	19.3
Block bootstrap	**9.2**	**8.7**	**15.3**	**27.2**	91.1	90.5	82.6	70.4
Mixed model	**7.4**	**8.0**	**13.6**	**24.5**	92.3	91.7	85.7	74.2
DM–HAC (time only)	**8.4**	**17.0**	**48.2**	**70.2**	91.8	82.4	51.8	28.1
Hering–Genton (space–time)	4.4	5.5	4.4	**2.6**	95.9	94.8	95.5	97.3
DA-K (kernel variant)	4.8	**7.2**	**9.9**	**12.8**	95.4	92.6	89.7	85.6
DA-P (proposed)	**3.1**	4.2	**3.4**	**2.4**	97.2	96.1	96.4	97.5
DA-P without $\hat{\tau}^2$ term	4.3	5.7	5.2	**3.4**	95.9	94.6	94.9	96.5
DA-P, normal reference	**3.3**	4.3	**3.4**	**3.0**	97.1	95.9	96.4	96.9
HAC, block-size bandwidth	6.3	5.6	**8.8**	**14.3**	94.1	94.3	90.5	83.6

Table 2 Experiment 1: rejection rate (%) of DA-P under the null according to whether the training variance was truncated ($\hat{\tau}^2_{\text{raw}} \leq 0$), and median Satterthwaite degrees of freedom.

		Rejection rate (%)		Median $\hat{\nu}$	
Dependence	Truncated (%)	Truncated	Not truncated	Truncated	Not truncated
None	71.3	4.1	0.7	806	15
Weak	62.8	6.1	1.1	1415	16
Moderate	64.9	4.6	1.1	299	26
Strong	75.7	2.5	2.1	41	28

training variance to $\sigma_\eta = 0.1$ and varied the true $\rho \in \{0, 0.25, 0.5, 0.75\}$, without and with moderate dependence (500 null replicates each), and ran DA-P with $\rho_0 \in \{0, 1/K, 0.5\}$ (Table 4). Under moderate dependence, DA-P stays at or below 5.6% whatever ρ and ρ_0, because the test-sampling variance dominates. Without spatio-temporal dependence, where the training variance is a large part of the total, DA-P is valid for $\rho \leq 0.25$ but rejects 10% of true nulls at $\rho = 0.5$ and 15% at $\rho = 0.75$ with the default $\rho_0 = 1/K$; assuming $\rho_0 = 0.5$ only reduces these rates to 8% and 13%. All other procedures are worse in these settings (14–29%). The common component of the fold effects, $\sqrt{\rho}\, g$ in our design, shifts all folds together and is therefore indistinguishable from a genuine difference in a single cross-validation run, as predicted by the result of Bengio and Grandvalet [13]. Highly correlated training effects arise when the training sets of different folds are nearly identical, i.e. with many folds, or when the learners are unstable in a way that depends on the whole sample; repeated cross-validation with different block allocations would provide information on ρ and is left for future work.

Table 3 Experiment 1: size-adjusted power (%) for true loss gaps $\delta \in \{0.1, 0.2, 0.4\}$ (300 replicates each). The critical value of each procedure is the 5% quantile of its own null p-values; it is undefined (–) when more than 5% of null bootstrap p-values equal the smallest attainable value.

	No dependence			Moderate			Strong		
Procedure	0.1	0.2	0.4	0.1	0.2	0.4	0.1	0.2	0.4
Paired t (folds)	24	54	96	7	13	29	5	7	9
i.i.d. bootstrap	31	82	100	–	–	–	–	–	–
Block bootstrap	29	81	100	9	18	46	9	12	17
Mixed model	30	81	100	10	20	49	5	8	12
DM–HAC (time only)	31	83	100	10	16	46	7	8	10
Hering–Genton (space–time)	25	63	89	9	17	46	5	8	12
DA-K (kernel variant)	26	61	89	7	14	29	5	7	12
DA-P (proposed)	22	56	83	8	16	46	5	9	11

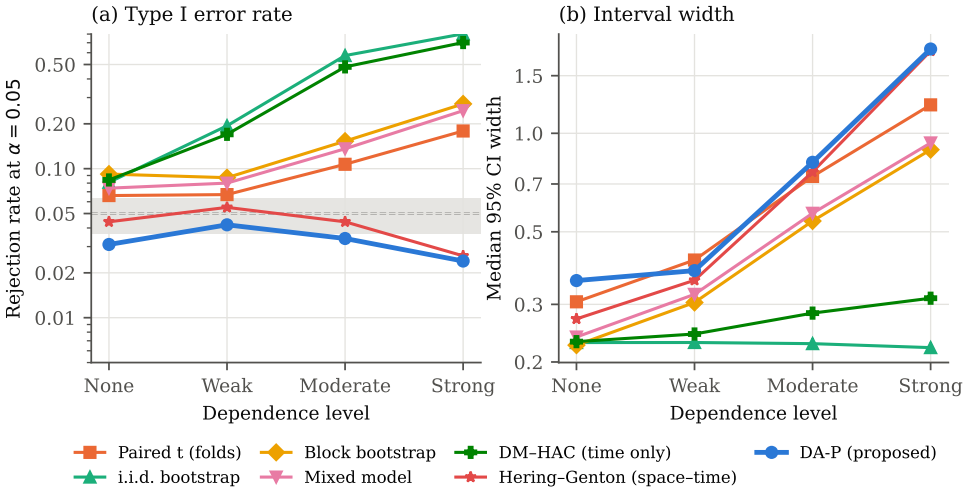


Fig. 2 Experiment 1. (a) Rejection rate of a true null hypothesis (log scale); the dashed line is the nominal level and the grey band the 95% Monte Carlo band. (b) Median width of the 95% confidence interval under the null.

5.4 Robustness to misspecification

DA-P assumes a separable, isotropic, stationary exponential correlation for the loss differential. Table 5 reports its behaviour, at the moderate dependence level, when this model is wrong: geometric anisotropy with ranges in ratio 4:1; a Gaussian covariance $\exp(-(h/a)^2)$, smoother than the exponential; heteroscedastic errors whose standard deviation varies by a factor of about 4.5 across the domain; a non-stationary field whose range increases from 1.5 to 12 grid units from one side of the domain to the other; and 144 sites placed uniformly at random instead of on a grid (500 null and 300 alternative replicates with $\delta = 0.2$ each). DA-P remains at or below the nominal

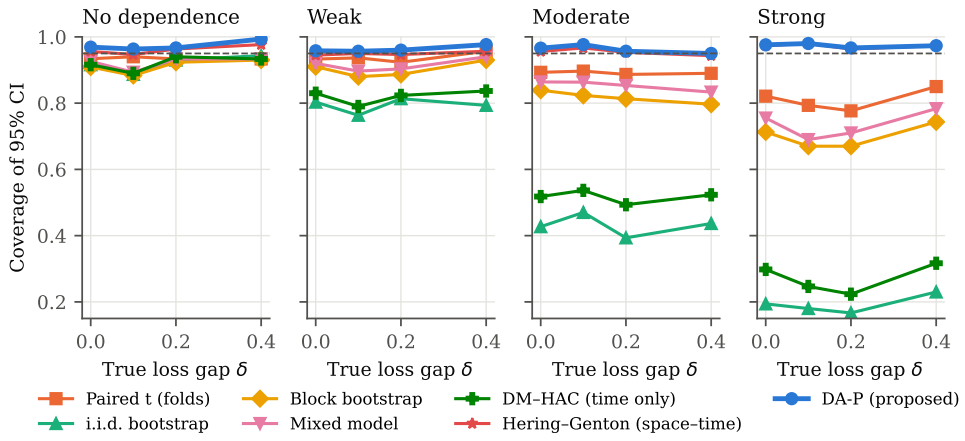


Fig. 3 Experiment 1. Coverage of the 95% confidence intervals for each true loss gap δ and dependence level (1000 replicates at $\delta = 0$, 300 otherwise). Dashed line: nominal coverage.

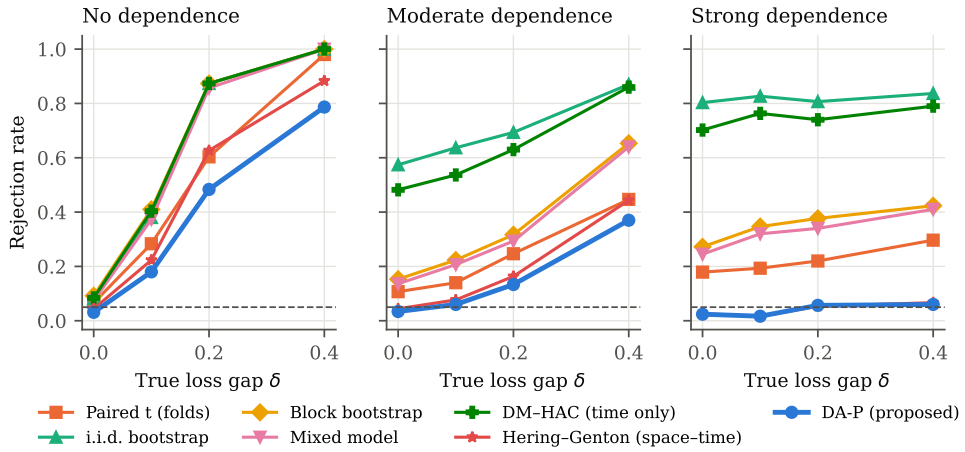


Fig. 4 Experiment 1. Raw rejection rate as a function of the true loss gap δ ($\delta = 0$ is the null). High curves at $\delta = 0$ indicate invalid tests, not power.

level in every scenario (0.8–4.6%), with coverage of 95–99%, as does Hering–Genton (0.8–6.2%), whereas the paired t-test, the cluster procedures and DA-K over-reject in all of them (7.6–19.6%). Misspecification costs power rather than validity: under the Gaussian covariance and heteroscedasticity, where the exponential model overstates the long-range correlation of the differentials, DA-P detects the gap in only 4% of replicates, against 16% for the (invalid) paired t-test.

Table 4 Sensitivity to the correlation of fold effects ($\sigma_\eta = 0.1$): rejection rate (%) of a true null (500 replicates per row) for the true correlation ρ and the assumed correlation ρ_0 of DA-P (default $\rho_0 = 1/K = 0.2$).

Dependence	ρ	Paired t	Block bs	LMM	H–G	DA-P, $\rho_0 = 0$	DA-P, $\rho_0 = 0.2$	DA-P, $\rho_0 = 0.5$
None	0	3.8	10.4	9.4	6.0	3.0	1.8	1.8
None	0.25	8.8	15.0	14.0	6.6	4.2	3.6	3.2
None	0.5	14.0	25.8	23.4	15.4	11.0	10.0	8.2
None	0.75	19.8	28.6	26.8	18.4	16.2	15.4	13.0
Moderate	0	9.2	14.4	12.0	2.6	2.2	2.2	2.0
Moderate	0.25	8.0	15.4	13.8	6.2	4.8	4.0	3.6
Moderate	0.5	10.8	18.6	16.0	6.0	5.6	4.6	3.8
Moderate	0.75	12.8	20.4	18.0	6.2	5.0	4.4	3.8

Table 5 Robustness at the moderate dependence level: rejection rate (%) of a true null (500 replicates) for six procedures, and coverage (%) and raw power (%) of DA-P and raw power of the paired t-test at $\delta = 0.2$ (300 replicates).

	Type I error (%)						DA-P, $\delta = 0.2$		Paired t
Scenario	Paired t	Block bs	LMM	H–G	DA-K	DA-P	Cov.	Power	Power
Baseline (moderate)	11.4	13.8	12.4	3.4	9.6	2.2	96.7	12	18
Anisotropic (4:1)	9.6	19.6	17.6	6.2	11.2	4.6	96.7	17	25
Gaussian covariance	9.2	11.6	9.2	0.8	7.6	0.8	99.3	4	16
Heteroscedastic	10.4	15.2	12.2	3.2	8.2	2.4	98.7	4	16
Non-stationary range	12.0	18.8	16.8	5.2	12.0	4.6	96.0	17	31
Irregular sites	7.8	14.2	12.2	3.2	9.0	2.8	95.3	12	18

5.5 Range of dependence and the decision rule

To examine how the procedures behave as the dependence range grows relative to the domain, and whether the range ratio can serve as a validity guard, we ran the design of this section with practical ranges $3a \in \{1.5, 3, 4.5, 6, 9, 12, 18\}$ grid units in space and the same values in time steps ($\phi = e^{-1/a}$), with 800 null replicates per range (300 from a pilot run and 500 from a confirmatory run on different seeds). Table 6 shows that DA-P remains at or below the nominal level over the whole range (1.8–3.4%), as does Hering–Genton (2.1–5.5%), whereas the rejection rates of the paired t-test (5.9% to 25.6%), of the block bootstrap (9.5% to 38.9%) and of DA-K (4.5% to 14.4%) increase steadily with the range. Figure 5 pools all null replicates of the simulation studies and plots the rejection rate against the estimated range ratio: in Experiment 1 designs the rejection rate of DA-P does not increase with the range ratio, and in Experiment 2 its residual over-rejection is concentrated at intermediate ratios (0.1–0.4) rather than at large ones. A threshold on the range ratio, which we had intended to use to declare comparisons indeterminate, would therefore not improve the validity of DA-P; large range ratios signal low power (the size-adjusted power of every procedure is low at the strong level, Table 3), not invalid inference.

The rule (12), which requires significance under both $\rho_0 = 1/K$ and $\rho_0 = 0.5$, is evaluated in Table 7 on the main replicates of both experiments. Its rate of false rejections is at most 4.0% in Experiment 1 and in the absence of or under weak

Table 6 Rejection rate (%) of a true null as a function of the practical range of dependence (800 replicates per row; domain of 12 sites per side and 24 time steps), with the median estimated range ratio of DA-P.

Range	Range ratio	n	Paired t	Block bs	H–G	DA-K	DA-P
1.5	0.11	800	5.9	9.5	4.5	4.5	2.6
3	0.15	800	7.0	10.0	5.5	7.6	3.4
4.5	0.21	800	6.9	11.6	4.5	7.9	3.2
6	0.28	800	9.2	13.2	3.5	9.2	2.6
9	0.40	800	12.9	20.8	3.0	10.4	3.1
12	0.52	800	17.2	27.4	2.5	12.6	2.8
18	0.75	800	25.6	38.9	2.1	14.4	1.8

Table 7 Three-way decision rule (12): percentage of replicates classified as "reject" (significant under $\rho_0 = 1/K$ and $\rho_0 = 0.5$) and as "indeterminate" (significant under $\rho_0 = 1/K$ only), under the null and under an alternative.

	Null (%)		Alternative (%)	
Dependence	Reject	Indeterminate	Reject	Indeterminate
Experiment 1 (alternative $\delta = 0.2$)				
None	2.9	0.2	42	7
Weak	4.0	0.2	38	4
Moderate	3.3	0.1	12	1
Strong	2.1	0.3	5	0
Experiment 2, blocked CV (alternative $\gamma = 0.2$)				
None	2.3	0.0	83	2
Weak	4.0	0.7	79	11
Moderate	8.7	1.3	51	11
Strong	8.7	2.3	29	9

dependence in Experiment 2, and 8.7% under moderate and strong dependence in Experiment 2, against 10.0% and 11.0% for DA-P alone. The price is that 0–11% of the comparisons with a genuine gap are classified as indeterminate instead of significant. The rule therefore reduces, but does not eliminate, false claims of superiority in the most difficult setting; we discuss why in Section 6.

6 Simulation study II: learners on spatio-temporal data

The second study reproduces the full pipeline of an applied comparison: data generation, model fitting by cross-validation and testing. The difficulty in such a study is to know the truth. We obtain an exact null by exchangeability.

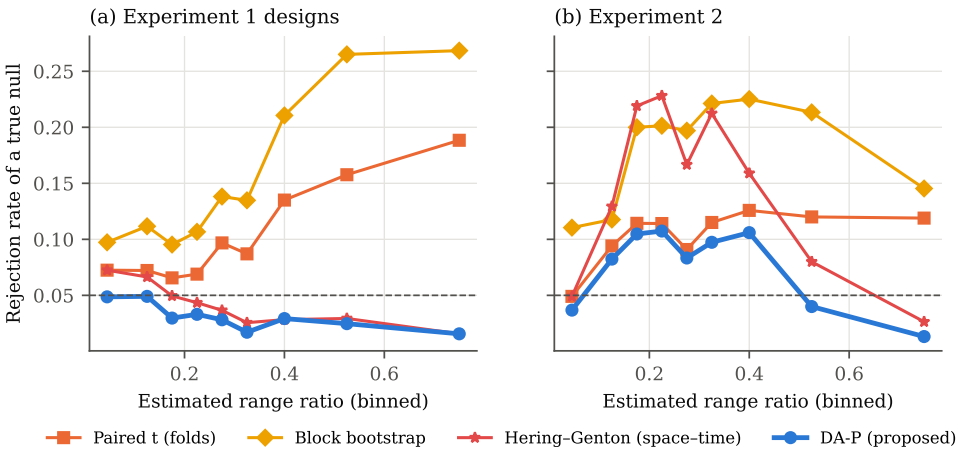


Fig. 5 Rejection rate of a true null against the estimated range ratio of DA-P, pooling all null replicates of (a) Experiment 1 and its range grid and (b) Experiment 2 (blocked CV). Bins with fewer than 40 replicates are omitted. Dashed line: nominal level.

6.1 Design

Data.

On a 10×10 grid observed at 20 time points ($n = 2000$), three covariates x_1, x_2, x_3 and a latent residual w are independent spatio-temporal fields with the dependence levels of Section 5. The response is

$$y = g(x_1) + (1-\gamma)\, g(x_2) + x_3 + w + \epsilon, \qquad g(v) = \sin(2v) + v, \qquad \epsilon \sim N(0, 0.5^2) \text{ i.i.d.}$$

Models.

Model A is XGBoost [41] (100 trees of depth 4, learning rate 0.1, row subsampling 0.8) fitted on (x_1, x_3, s, t); model B is the same learner fitted on (x_2, x_3, s, t). Including the coordinates lets both models exploit the spatial and temporal structure of w, as is common in practice.

Exact null by exchangeability.

When $\gamma = 0$, the joint distribution of the data is invariant under the exchange of x_1 and x_2, which exchanges the roles of A and B. Hence $\mathbb{E}[d_i] = 0$ for every observation and every validation design: the two models have exactly the same expected performance, although in any given sample one of them will appear better. When $\gamma > 0$, x_1 carries more signal than x_2 and A is truly better; we use $\gamma \in \{0.1, 0.2, 0.3\}$.

Validation and procedures.

Two pipelines are applied to each data set. The *conventional* pipeline uses random 5-fold cross-validation with the paired t-test on fold means and the i.i.d. bootstrap, and

5×5 repeated random cross-validation with the corrected resampled t-test [2]. The *dependence-aware* pipeline uses blocked 5-fold cross-validation (5×5-site tiles crossed with 4-step windows, $G = 20$ blocks, and a buffer that removes from the training set observations within one site and one time step of the test fold) with the procedures of Section 5, including DA-P with $\rho_0 = 0$ and $\rho_0 = 0.5$. There are 300 replicates per dependence level under the null and 150 under each alternative.

6.2 Results

Random cross-validation.

The left panel of Fig. 6 and the upper part of Table 8 show that the conventional pipeline fails as soon as the data are dependent. With random 5-fold cross-validation, the paired t-test rejects a true null in 15% of replicates under moderate and 29% under strong dependence, and the i.i.d. bootstrap in 24% and 40%. The corrected resampled t-test of Nadeau and Bengio is well calibrated without dependence (3.7%) and under weak dependence (5.7%), as designed, but not beyond (16% and 33%): its correction accounts for overlapping training sets, not for dependent test observations. Note also that even without spatio-temporal dependence the paired t-test rejects 9% of true nulls, the problem identified by Dietterich [1].

The estimand changes with the validation design.

Random cross-validation does not only produce invalid tests; it also estimates a different quantity. Table 9 reports the average estimated gain of model A under both designs. Because both models can exploit the coordinates to interpolate between nearby training observations, random cross-validation attenuates the advantage conferred by the more informative covariate, by about 30% under moderate and 50% under strong dependence (-0.53 instead of -0.80, and -0.30 instead of -0.60, for $\gamma = 0.2$). A gain measured by random cross-validation is thus not a conservative estimate of the gain at new locations and times: it is an estimate of a different, interpolation-based, quantity.

Blocked cross-validation.

Blocking is necessary but not sufficient. With blocked folds, procedures that assume independent observations (i.i.d. bootstrap, 55% at the moderate level) or ignore cross-sectional correlation (DM–HAC, 50%) are even worse than under random cross-validation, because the dependent test errors are no longer diluted across folds. The cluster procedures over-reject at every level, including without dependence (11% and 8%), since the fold effects induced by refitting XGBoost are correlated across blocks of the same fold. The Hering–Genton test, valid in Experiment 1, now rejects 17% of true nulls under moderate and 23% under strong dependence: it models the spatial correlation of the differentials but not the training-induced component, which is large in this design. DA-P is the best procedure at every level (2.3%, 4.7%, 10.0%, 11.0%), ahead of the paired t-test (5.3%, 7.7%, 14.3%, 14.3%) and DA-K (3.3%, 7.0%, 15.0%, 14.3%), but it is not calibrated under moderate and strong dependence.

The residual over-rejection has a clear source. Whichever of the two covariate fields happens to be more variable over the whole domain in a given sample favours the

Table 8 Experiment 2: rejection rate (%) of a true null ($\gamma = 0$, 300 replicates per level) and raw power (%) at the moderate dependence level (150 replicates per value of γ). Bold: rejection rate outside the 95% Monte Carlo band [2.5, 7.5].

		Type I error (%)				Power, moderate (%)		
CV	Procedure	None	Weak	Mod.	Strong	$\gamma = 0.1$	0.2	0.3
Random	Paired t (folds)	**9.0**	**8.3**	**14.7**	**29.3**	67.3	99.3	100.0
Random	i.i.d. bootstrap	7.0	**12.7**	**23.7**	**40.0**	80.0	99.3	100.0
Random	Corrected resampled t	3.7	5.7	**16.3**	**32.7**	71.3	100.0	100.0
Blocked	Paired t (folds)	5.3	**7.7**	**14.3**	**14.3**	24.7	66.7	89.3
Blocked	i.i.d. bootstrap	5.3	**22.7**	**55.0**	**66.3**	77.3	96.0	99.3
Blocked	Block bootstrap	**10.7**	**12.3**	**20.3**	**27.7**	42.7	80.7	98.0
Blocked	Mixed model	**8.0**	**10.3**	**18.0**	**21.3**	38.0	76.0	97.3
Blocked	DM–HAC (time only)	6.3	**20.7**	**49.7**	**59.3**	72.0	93.3	99.3
Blocked	Hering–Genton (space–time)	3.3	**10.0**	**16.7**	**22.7**	41.3	75.3	97.3
Blocked	DA-K (kernel variant)	3.3	7.0	**15.0**	**14.3**	22.7	62.0	80.0
Blocked	DA-P (proposed)	**2.3**	4.7	**10.0**	**11.0**	24.0	62.7	84.0

Table 9 Experiment 2: Monte Carlo mean of the estimated loss differential $\hat{\Delta}$ (model A minus model B, squared loss) under random (R) and blocked (B) cross-validation. Under the null ($\gamma = 0$) both are zero by exchangeability.

	$\gamma = 0$		$\gamma = 0.1$		$\gamma = 0.2$		$\gamma = 0.3$	
Dependence	R	B	R	B	R	B	R	B
None	-0.003	0.002	-0.395	-0.401	-0.809	-0.827	-1.114	-1.133
Weak	-0.012	-0.013	-0.356	-0.425	-0.699	-0.821	-1.022	-1.205
Moderate	-0.009	-0.031	-0.275	-0.403	-0.528	-0.797	-0.747	-1.145
Strong	-0.000	0.020	-0.155	-0.293	-0.297	-0.597	-0.424	-0.847

corresponding model in every fold; with practical ranges of a quarter to a half of the domain (median range ratio 0.27 and 0.36), this sample-wide component acts as a fold effect that is common to all folds, i.e. as a large correlation ρ between training effects. By Proposition 2, it is then underestimated, and by the result of Bengio and Grandvalet [13] it cannot be estimated from a single cross-validation run. Consistently, assuming $\rho_0 = 0.5$ lowers the rejection rate to 8.7% at both levels, and assuming $\rho_0 = 0$ raises it to 11.7% and 15.3%; the three-way rule of Section 4.7, evaluated in Table 7, inherits the 8.7%.

Power.

Under the alternatives (right part of Table 8), DA-P and the blocked paired t-test have similar power at the moderate level (63% and 67% for $\gamma = 0.2$), lower than that of the invalid procedures. Without dependence, DA-P is less powerful than the paired t-test for large effects (80% against 100% for $\gamma = 0.3$), because a large fold-level variance $\hat{\tau}^2$, estimated with $K - 1$ degrees of freedom, then dominates its variance and reduces its degrees of freedom.

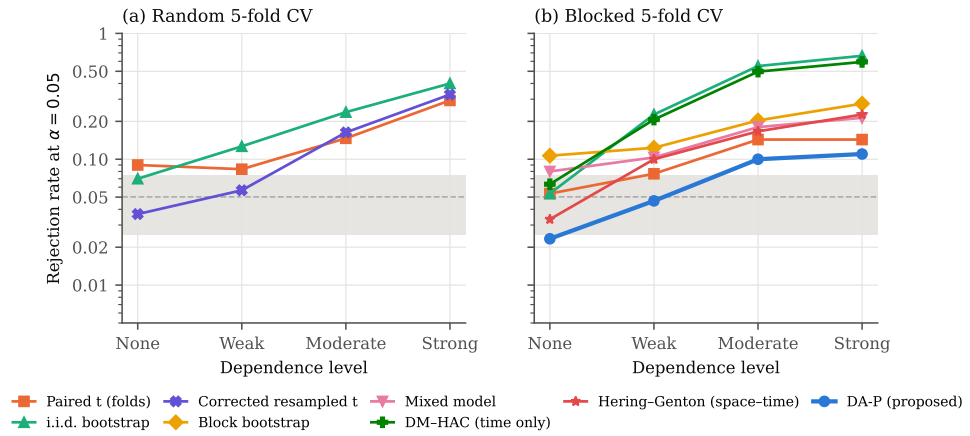


Fig. 6 Experiment 2. Rejection rate of a true null hypothesis (two XGBoost models with exchangeable feature sets) under random and blocked cross-validation (log scale). Dashed line: nominal level; grey band: 95% Monte Carlo band.

7 Case studies

7.1 Data and protocol

We use three public data sets obtained from the Rdatasets collection [42].

NYC weather.

Hourly weather records at the three New York City airports (EWR, JFK, LGA) in 2013, distributed with the `nycflights13` package [43] ($n = 26{,}110$ after removing five records with missing temperature, humidity or wind speed). The task is to classify whether precipitation was recorded in the hour, from temperature, dew point, humidity, wind direction and speed, pressure, month, hour and airport; 6.7% of hours are positive. The loss is the logarithmic loss.

Melbourne pedestrians.

Hourly pedestrian counts at four sensors in Melbourne in 2015–2016, distributed with the `tsibble` package [44] ($n = 66{,}037$). The task is to predict $\log(1 + \text{count})$ from hour, day of week, month, day of year, year and sensor, under squared loss.

Las Rosas maize.

Yield-monitor data from a maize field in Las Rosas, Argentina, in 1999 and 2001 [45], distributed with the `agridat` package ($n = 3443$, field of about 660×380 m). The task is to predict yield from the nitrogen rate, brightness value, topographic zone, year and coordinates, under squared loss.

For the two hourly data sets, blocks are two-week windows (26 and 53 blocks); as the airports lie within about 35 km of one another and the sensors within about

2 km, they are treated as co-located, so that all sites observed at the same time may be correlated, and a 24-hour buffer is removed from the training set on each side of a test block. For the maize data, blocks are 75×75 m tiles (35 non-empty blocks) with a 10 m buffer. We compare three learners pairwise: XGBoost (300 trees of depth 6, learning rate 0.05, row and column subsampling 0.8), a random forest [46] (200 trees, minimum leaf size 5, 50% of features per split) and a linear baseline (logistic or ridge regression on standardised, mean-imputed features). All models use scikit-learn [47] or the XGBoost library with default settings otherwise; no hyperparameter was tuned, since our object is the comparison, not the models.

Interpolation and forecasting.

Blocked cross-validation with blocks distributed over the whole period predicts each held-out window from data both before and after it; it answers an *interpolation* question (filling gaps in a known record). For the two hourly data sets we therefore also use *forward chaining* [48]: the 26 or 53 two-week blocks are grouped into $K = 5$ consecutive segments, and each of the last four segments is predicted from all observations that precede it, minus a 24-hour buffer, which answers a *forecasting* question. The first segment is used for training only, the training sets are nested, and the same tests are applied to the $K - 1 = 4$ test segments.

Multiplicity.

With three learners, each data set and validation design involves three pairwise comparisons. We report raw p-values and p-values adjusted by Holm's procedure [49] within each data set, validation design and test, and we treat the case studies as illustrations of how conclusions depend on the inferential choices rather than as definitive rankings of the learners.

Seed-null diagnostic.

For each data set we also compare XGBoost with itself, the two runs differing only in their random seed (30 pairs). By symmetry, the expected loss differential of such a comparison is zero, while in each run the stochastic subsampling produces a non-zero observed difference. A procedure that frequently declares one seed "significantly better" than the other confuses training noise with a real improvement. Because the data are held fixed, this diagnostic measures the sensitivity of each procedure to training randomness only; it is not a full assessment of size, which Sections 5 and 6 provide.

7.2 Results

Table 10 and Fig. 7 report the estimated loss differentials (first learner minus second; negative values favour the first), their 95% intervals, and the p-values of the main procedures. Table 11 gives the DA-P diagnostics and the outcome of the three-way rule (12), and Fig. 8 the variograms of the fold-centred loss differentials.

NYC weather: the ranking depends on the question.

Under random cross-validation, XGBoost appears clearly better than the random forest (log-loss gain 0.011) and than logistic regression (0.041), and the random forest

better than logistic regression (0.030), with $p \leq 0.001$ for every procedure. Under two-week blocked cross-validation, i.e. for filling gaps of two weeks in the record, the three differences vanish ($|\hat{\Delta}| \leq 0.004$, DA-P $p \geq 0.43$): random folds let the models recognise weather episodes from neighbouring hours of the same day, and the apparent superiority of XGBoost reflects its better use of this leakage. The i.i.d. bootstrap applied to the blocked differentials even declares the random forest significantly *better* than XGBoost ($p = 0.023$), a reversal produced by ignoring the 20-hour dependence range of the loss differential. Under forward chaining, i.e. for forecasting later weeks from earlier ones, the random forest is better than XGBoost by 0.030 (DA-P $p = 0.008$, Holm-adjusted 0.023, significant also with $\rho_0 = 0.5$), while the paired t-test, with three degrees of freedom, does not detect this difference ($p = 0.050$, adjusted 0.15). For XGBoost against logistic regression under forward chaining, the Hering–Genton test reports $p = 0.036$ whereas DA-P, whose training term dominates the variance ($\hat{\nu} = 4$), reports $p = 0.42$: with nested training sets of very different sizes, the fold-to-fold variability of the comparison is large, and only a procedure that accounts for it avoids a spurious conclusion.

Melbourne pedestrians: a small gain of uncertain reality.

XGBoost and the random forest are indistinguishable under random cross-validation and under forward chaining, but under blocked cross-validation XGBoost has a small advantage (0.0025 in squared error of the log count). The paired t-test ($p = 0.033$) and the i.i.d. bootstrap ($p = 0.011$) call it significant; DA-P and Hering–Genton ($p = 0.47$) and the cluster procedures ($p \geq 0.40$) do not. The variogram of the loss differential has a short estimated range (5 hours), but pedestrian traffic also has daily and weekly cycles and holiday effects that an exponential model does not represent; we would conclude that the gain is not established. By contrast, the advantage of both tree ensembles over the linear baseline (1.85) is overwhelming under every procedure and validation design.

Las Rosas maize: limited information.

The gain of XGBoost over the random forest is $2.8\,(\mathrm{q/ha})^2$ under random cross-validation, significant with all procedures ($p \leq 0.013$), and much larger but much more uncertain under spatial blocking (21). The i.i.d. bootstrap ($p = 0.001$) and the block bootstrap ($p = 0.043$) declare it significant; the paired t-test ($p = 0.31$), Hering–Genton ($p = 0.51$) and DA-P ($p = 0.46$) do not. The DA-P diagnostics explain why: the loss differential has a spatial range of about 150 m in a field of 660×380 m (range ratio 0.40), so that the 3443 yield-monitor points carry the information of only about 20 independent observations. With so little information, even the large gains over the linear model are only moderately significant: $p = 0.011$ (adjusted 0.032) for XGBoost, which survives the three-way rule, and $p = 0.043$ (adjusted 0.086) for the random forest, which does not survive the correction for multiplicity. Under random cross-validation all of these comparisons appear decisive.

Table 10 Case studies: estimated loss differential $\hat{\Delta}$ (first minus second learner; negative favours the first) with 95% confidence interval, and p-values of five procedures (for R, the paired t-test is computed on one random 5-fold run); the last column is the DA-P p-value adjusted by Holm's procedure over the three comparisons of each data set and validation design. R: random CV (estimate and interval from the corrected resampled t-test on 5×5 repeated CV); B: blocked 5-fold CV and F: forward chaining (estimate and interval from DA-P). XGB: XGBoost; RF: random forest; Lin: linear baseline. Losses: log loss (NYC weather), squared error of $\log(1 + \text{count})$ (pedestrians), squared error of yield in $(\text{q/ha})^2$ (maize). Detailed results for all procedures are given in Appendix A.

			p-value					
Comp.	CV	$\hat{\Delta}$ [95% CI]	Paired t	i.i.d. bs	Block bs	H–G	DA-P	Holm
NYC weather (rain, log loss)								
XGB–RF	R	-0.0115 [-0.0131, -0.0098]	<0.001	0.001	–	–	–	–
	B	0.0036 [-0.0055, 0.0127]	0.454	0.023	0.279	0.435	0.438	1.000
	F	0.0299 [0.0080, 0.0519]	0.050	0.001	0.002	0.007	0.008	0.023
XGB–Lin	R	-0.0413 [-0.0453, -0.0373]	<0.001	0.001	–	–	–	–
	B	0.0028 [-0.0109, 0.0165]	0.459	0.137	0.466	0.685	0.685	1.000
	F	0.0256 [-0.0511, 0.1023]	0.308	0.001	0.050	0.036	0.418	0.837
RF–Lin	R	-0.0302 [-0.0338, -0.0265]	<0.001	0.001	–	–	–	–
	B	-0.0005 [-0.0109, 0.0099]	0.978	0.763	0.875	0.928	0.928	1.000
	F	-0.0064 [-0.0777, 0.0649]	0.717	0.001	0.438	0.271	0.803	0.837
Melbourne pedestrians (log count, MSE)								
XGB–RF	R	0.0003 [-0.0020, 0.0027]	0.795	0.757	–	–	–	–
	B	-0.0025 [-0.0093, 0.0043]	0.033	0.011	0.402	0.473	0.473	0.473
	F	-0.0003 [-0.0099, 0.0092]	0.855	0.787	0.932	0.947	0.947	0.947
XGB–Lin	R	-1.89 [-1.91, -1.87]	<0.001	0.001	–	–	–	–
	B	-1.85 [-1.91, -1.79]	<0.001	0.001	0.001	<0.001	<0.001	<0.001
	F	-1.83 [-2.26, -1.41]	<0.001	0.001	0.001	<0.001	<0.001	0.001
RF–Lin	R	-1.89 [-1.91, -1.87]	<0.001	0.001	–	–	–	–
	B	-1.85 [-1.91, -1.78]	<0.001	0.001	0.001	<0.001	<0.001	<0.001
	F	-1.83 [-2.27, -1.40]	<0.001	0.001	0.001	<0.001	<0.001	0.001
Las Rosas maize (yield, MSE)								
XGB–RF	R	-2.36 [-3.53, -1.19]	0.013	0.001	–	–	–	–
	B	-21.1 [-80.7, 38.4]	0.308	0.001	0.043	0.511	0.459	0.459
XGB–Lin	R	-131.7 [-137.5, -125.9]	<0.001	0.001	–	–	–	–
	B	-160.0 [-277.1, -42.8]	<0.001	0.001	0.001	0.002	0.011	0.032
RF–Lin	R	-129.4 [-135.1, -123.6]	<0.001	0.001	–	–	–	–
	B	-133.2 [-261.5, -4.8]	0.018	0.001	0.001	0.047	0.043	0.086

Seed-null diagnostic.

Table 12 reports how often each procedure declares one of two XGBoost runs, differing only by their seed, significantly better than the other. Procedures that resample observations independently do so in 33–87% of the pairs under blocked cross-validation, and in 40% of the pairs for the pedestrian data under random cross-validation, where the subsampling noise of the learner is temporally coherent. DM–HAC does so in 23–33% and Hering–Genton in 20% of the pedestrian pairs. DA-P (0–3%) and the cluster procedures (0–13%) are the least sensitive to training noise, followed by the blocked paired t-test (0–13%). With 30 pairs per data set, these rates have a Monte Carlo standard error of up to 9 percentage points and should be read qualitatively.

Table 11 Case studies: DA-P diagnostics (estimated practical ranges, effective sample size $\hat{n}_{\text{eff}}$, degrees of freedom $\hat{\nu}$, range ratio) and outcome of the three-way rule (12) on raw p-values. B: blocked CV; F: forward chaining.

Data set	Comparison	CV	$\hat{r}_S$	$\hat{r}_T$	$\hat{n}_{\text{eff}}$	$\hat{\nu}$	$\hat{r}/L$	Decision
NYC weather	XGBoost vs random forest	B	–	20 h	2765	666	0.00	not rej.
NYC weather	XGBoost vs random forest	F	–	34 h	886	292	0.01	reject
NYC weather	XGBoost vs linear	B	–	7 h	1830	1829	0.00	not rej.
NYC weather	XGBoost vs linear	F	–	34 h	1060	4	0.01	not rej.
NYC weather	Random forest vs linear	B	–	5 h	2724	2723	0.00	not rej.
NYC weather	Random forest vs linear	F	–	75 h	1439	3	0.01	not rej.
Melbourne pedestrians	XGBoost vs random forest	B	–	5 h	4819	4818	0.00	not rej.
Melbourne pedestrians	XGBoost vs random forest	F	–	6 h	3442	3441	0.00	not rej.
Melbourne pedestrians	XGBoost vs linear	B	–	38 h	6008	689	0.00	reject
Melbourne pedestrians	XGBoost vs linear	F	–	7 h	2984	4	0.00	reject
Melbourne pedestrians	Random forest vs linear	B	–	41 h	5910	648	0.00	reject
Melbourne pedestrians	Random forest vs linear	F	–	7 h	2982	4	0.00	reject
Las Rosas maize	XGBoost vs random forest	B	153 m	–	19	14	0.40	not rej.
Las Rosas maize	XGBoost vs linear	B	171 m	–	17	16	0.45	reject
Las Rosas maize	Random forest vs linear	B	146 m	–	19	18	0.38	reject

Table 12 Seed-null diagnostic: percentage of 30 pairs of XGBoost runs differing only in their random seed for which the procedure rejects equality at $\alpha = 0.05$. –: not available (no time axis).

	Random CV		Blocked CV						
Data set	Paired t	i.i.d. bs	Paired t	i.i.d. bs	Block bs	LMM	DM–HAC	H–G	DA-P
NYC weather	0	7	13	33	13	10	23	0	0
Melbourne pedestrians	0	40	7	63	0	0	33	20	3
Las Rosas maize	3	7	0	87	0	0	–	0	0

8 Discussion

8.1 Are machine learning improvements statistically real?

Our results give a nuanced answer to the question in the title. With spatially or temporally dependent data, the conventional pipeline (random K-fold cross-validation followed by a paired t-test, an i.i.d. bootstrap, or even the corrected resampled t-test) produces false discoveries at rates between 15% and 40% under moderate to strong dependence in our simulations, instead of the nominal 5%. Moreover, random cross-validation targets an interpolation estimand that can understate the gain at new places and times by half, and the case studies show that the ranking of the same learners can change with the prediction task: on hourly weather data, XGBoost appears best under random folds, the learners are indistinguishable when filling two-week gaps, and the random forest is significantly better when forecasting later weeks. Many reported "significant" improvements on such data are therefore not statistically supported, and some are answers to a different question from the one of interest.

Switching to blocked cross-validation fixes the estimand but not, by itself, the inference: applied to blocked folds, procedures that resample observations or that model only temporal dependence are *more* anti-conservative than under random folds (60–80% false positives under strong dependence). Cluster procedures (block bootstrap,

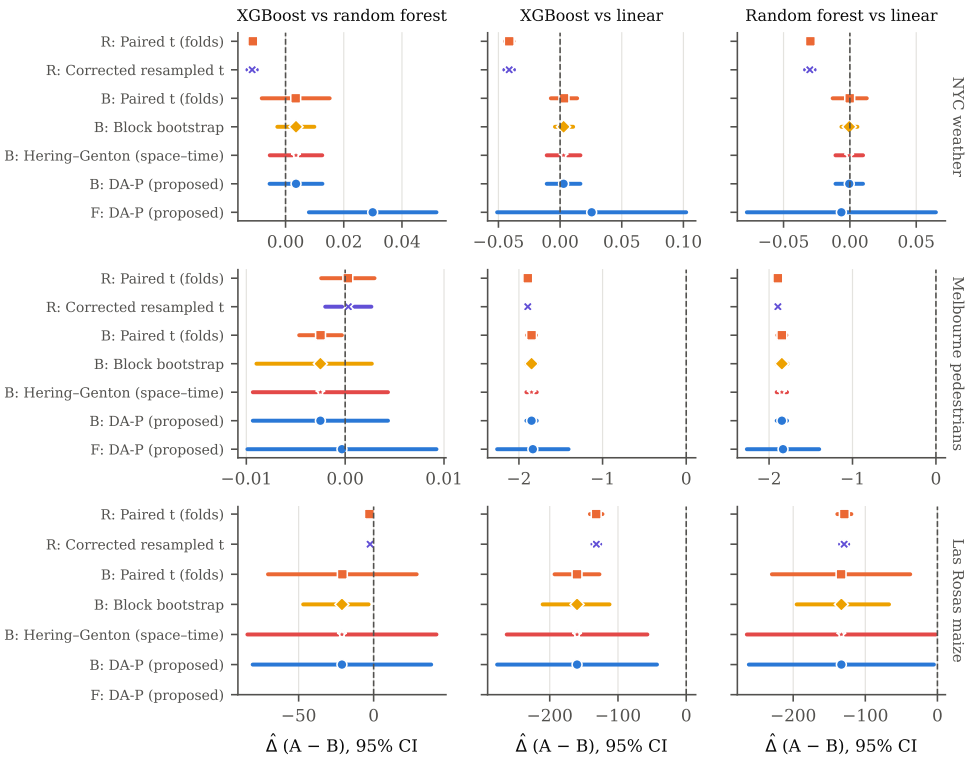


Fig. 7 Case studies: 95% confidence intervals for the loss differential under random (R), blocked (B) and forward-chaining (F) cross-validation. The dashed line marks equal performance.

random-intercept models) remain anti-conservative whenever the dependence range is not small compared with the blocks, and they ignore the correlation of training-induced fold effects. The blocked paired t-test, whose behaviour is explained by Proposition 1, is the best conventional procedure but still rejects 11–18% of true nulls under moderate and strong dependence.

8.2 What DA-P adds, and what it does not

Relative to the Hering–Genton test.

The closest existing procedure is the spatial prediction comparison test of Hering and Genton [10], which we extended to space–time. When the training-induced variability is small (Experiment 1 and its variants), both procedures are valid, and Hering–Genton is somewhat more powerful because DA-P spends degrees of freedom on a training term that is nearly zero. When the models are refitted learners with substantial fold-to-fold variability (Experiment 2), Hering–Genton rejects 17–23% of true nulls under moderate and strong dependence, against 10–11% for DA-P; on real data, it declares one

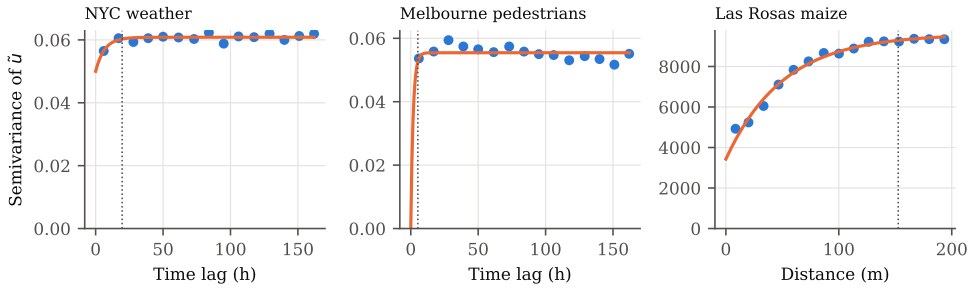


Fig. 8 Case studies: empirical semivariograms of the fold-centred loss differentials $\tilde{u}$ (XGBoost vs random forest, blocked CV; points), fitted exponential models (lines) and practical ranges (dotted).

seed of XGBoost significantly better than another in 20% of the pedestrian pairs (DA-P: 3%), and reports a significant forecasting difference between XGBoost and logistic regression that DA-P attributes to fold-to-fold variability. The training term, whose estimation and limits are characterised by Proposition 2, is thus what distinguishes the comparison of *learning procedures* from the comparison of fixed predictions.

Relative to the kernel variant.

Our initial kernel version, DA-K, over-rejected under moderate and strong dependence (10–15%); the pilot study traced this to replicates in which the training variance was truncated at zero, where the downward bias of the fold-centred kernel estimate was no longer compensated. The parametric variance of DA-P does not suffer from kernel down-weighting, and it remained valid under anisotropy, a smoother covariance, heteroscedasticity, non-stationary ranges and irregular sampling, misspecification costing power rather than validity.

What remains.

DA-P is not exactly calibrated in the most difficult setting we studied: learners compared under moderate or strong dependence, where it rejects 10–11% of true nulls (8.7% with the three-way rule). Proposition 2 identifies the reason: whichever covariate field happens to be more informative over the whole sample favours one model in every fold, which acts as a fold effect common to all folds, i.e. a training correlation ρ larger than assumed, and such a component cannot be estimated from a single cross-validation run [13]. The same mechanism explains the over-rejection of all procedures when fold effects are highly correlated in Section 5.3. The range ratio, which we initially intended to use as a guard, does not identify these cases (Section 5.5); it is a useful indicator of how much information a comparison can carry, not of its validity.

8.3 Recommendations

From these results we derive the following recommendations for comparing two models on spatio-temporal data.

1. *Define the estimand first.* Choose the validation design to match the intended use of the model: random cross-validation answers an interpolation question at the scale of single observations, blocked cross-validation an interpolation question at the scale of the blocks, and forward chaining a forecasting question.
2. *Never rely on procedures that treat observations as independent* (i.i.d. bootstrap, z- or t-tests on pooled observations, McNemar's test) or on the paired t-test with random folds. Under dependence, the corrected resampled t-test is not a remedy either.
3. *Use blocked cross-validation with a test that accounts for both sources of dependence*, such as DA-P, and report its result under $\rho_0 = 1/K$ and $\rho_0 = 0.5$; claim an improvement only when it holds under both (the three-way rule), and report the other cases as indeterminate.
4. *Inspect the diagnostics.* Report the estimated ranges, the effective sample size and the range ratio. A small effective sample (a few dozen, as on the Las Rosas field) means that only large differences can be resolved.
5. *Correct for multiplicity* when several models are compared, and treat comparisons on a single data set as exploratory.
6. *Report intervals, not only p-values*, on the scale of the loss, as advocated for large data sets in this journal [35]: an improvement can be statistically real and practically negligible, or practically important and statistically unresolved.

8.4 Limitations and future work

Our simulation designs are stylised: Gaussian fields on a 12×12 or 10×10 grid (with one irregular variant), a single learner family in Experiment 2, and $K = 5$ folds throughout. The working model (2) assumes additive fold effects and a stationary loss differential, which may fail when one model is better in some regions and worse in others; a version with a spatially varying Δ is a natural extension. The correlation ρ of training effects is assumed rather than estimated; repeated blocked cross-validation with different block allocations, or nested cross-validation in the spirit of Bates et al. [16], could provide information on it and is, in our view, the most promising route to a fully calibrated test. The exponential variogram does not represent periodic dependence such as daily and weekly cycles, which matter for the pedestrian data. We considered additive losses only; non-additive metrics such as the AUC or the F1-score would require linearisation. Finally, the choice of the parametric variance and of the three-way rule was informed by pilot studies on separate seeds and, for the rule, by the failure of a range-based alternative on the main replicates; both deserve confirmation on further designs and data sets.

9 Conclusion

Whether a machine learning model is "better" than another is a statistical claim, and on spatio-temporal data it depends on two choices that are rarely made explicit: the validation design, which defines what is estimated, and the inferential procedure, which defines how its uncertainty is assessed. We formalised the comparison of two learning procedures as inference on the mean of a dependent loss-differential field

with correlated fold-level training effects, derived the bias of the paired t-test and of a combined variance estimator, and proposed DA-P, a test that adds to a variogram-based test-sampling variance a training-variance term and a Satterthwaite reference distribution, all computed from one blocked cross-validation run, together with a three-way decision rule that makes the dependence of the conclusion on the unverifiable training correlation explicit.

In controlled simulations, conventional procedures declared false improvements up to sixteen times more often than their nominal level, and random cross-validation could halve the estimated gain of a genuinely better model. DA-P kept the error rate at or below the nominal level in all designs with fixed error fields and weakly correlated training effects, including under anisotropy, a different covariance shape, heteroscedasticity, non-stationarity and irregular sampling, and it was the best procedure with refitted gradient-boosting learners, where the Hering–Genton test, designed for fixed predictions, failed. It was not exactly calibrated in that last setting under moderate and strong dependence, because a training component shared by all folds cannot be identified from a single cross-validation run, which also causes over-rejection when training effects are strongly correlated; we have characterised this limit rather than hidden it. On three public data sets, the conclusion of six of the nine comparisons depended on the validation design or on the test, and on hourly weather data the ranking of XGBoost and the random forest changed with the prediction task; the large improvements of tree ensembles over linear baselines on the hourly data were confirmed by every procedure.

Trustworthy machine learning requires not only accurate and explainable models but also trustworthy evidence that one model is better than another. We hope that the procedure, its diagnostics and the open-source implementation presented here will help practitioners in environmental, agricultural, epidemiological and mobility applications to report improvements that are statistically real, and to say so when the data cannot tell.

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Declarations

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- **Conflict of interest.** The author declares no competing interests.
- **Ethics approval and consent to participate.** Not applicable.
- **Consent for publication.** Not applicable.
- **Data availability.** All data sets are public and are redistributed, with their original licences, in the code repository, together with the raw outputs of all experiments.
- **Code availability.** The Python package `dacomp` and all scripts needed to reproduce the simulations, case studies, tables and figures are available at https://github.com/2zalab/Statistics-for-Data-Science-and-AI under the MIT licence.

- **Author contribution.** I.T. conceived the study, developed the methodology and software, carried out the experiments and wrote the manuscript.

Appendix A Detailed results of the case studies

Tables A1–A3 report, for each data set, comparison and validation design, the estimated loss differential with the 95% interval of each procedure, its raw p-value and the p-value adjusted by Holm's procedure over the three comparisons of the data set (same validation design and procedure). R: random CV; B: blocked CV; F: forward chaining.

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Table A1 Case study: NYC weather (log loss).

CV	Procedure	$\hat{\Delta}$ [95% CI]	p	p_{Holm}
XGBoost vs random forest				
R	Paired t (folds)	-0.0112 [-0.0121, -0.0103]	<0.001	<0.001
R	i.i.d. bootstrap	-0.0112 [-0.0131, -0.0096]	0.001	0.003
R	Corrected resampled t	-0.0115 [-0.0131, -0.0098]	<0.001	<0.001
B	Paired t (folds)	0.0035 [-0.0082, 0.0152]	0.454	1.000
B	i.i.d. bootstrap	0.0036 [0.0005, 0.0067]	0.023	0.069
B	Block bootstrap	0.0036 [-0.0028, 0.0099]	0.279	0.837
B	Mixed model	0.0036 [-0.0033, 0.0105]	0.291	0.874
B	DM–HAC (time only)	0.0036 [-0.0007, 0.0079]	0.099	0.298
B	Hering–Genton (space–time)	0.0036 [-0.0054, 0.0126]	0.435	1.000
B	DA-K (kernel variant)	0.0036 [-0.0076, 0.0148]	0.499	1.000
B	DA-P (proposed)	0.0036 [-0.0055, 0.0127]	0.438	1.000
F	Paired t (folds)	0.0300 [-0.0001, 0.0601]	0.050	0.151
F	Block bootstrap	0.0299 [0.0132, 0.0500]	0.002	0.006
F	Hering–Genton (space–time)	0.0299 [0.0081, 0.0517]	0.007	0.021
F	DA-P (proposed)	0.0299 [0.0080, 0.0519]	0.008	0.023
XGBoost vs linear				
R	Paired t (folds)	-0.0413 [-0.0447, -0.0378]	<0.001	<0.001
R	i.i.d. bootstrap	-0.0413 [-0.0449, -0.0380]	0.001	0.003
R	Corrected resampled t	-0.0413 [-0.0453, -0.0373]	<0.001	<0.001
B	Paired t (folds)	0.0032 [-0.0075, 0.0139]	0.459	1.000
B	i.i.d. bootstrap	0.0028 [-0.0010, 0.0065]	0.137	0.274
B	Block bootstrap	0.0028 [-0.0042, 0.0106]	0.466	0.932
B	Mixed model	0.0028 [-0.0054, 0.0111]	0.487	0.975
B	DM–HAC (time only)	0.0028 [-0.0030, 0.0087]	0.344	0.687
B	Hering–Genton (space–time)	0.0028 [-0.0109, 0.0165]	0.685	1.000
B	DA-K (kernel variant)	0.0028 [-0.0052, 0.0109]	0.491	1.000
B	DA-P (proposed)	0.0028 [-0.0109, 0.0165]	0.685	1.000
F	Paired t (folds)	0.0256 [-0.0409, 0.0922]	0.308	0.616
F	Block bootstrap	0.0256 [0.0005, 0.0513]	0.050	0.100
F	Hering–Genton (space–time)	0.0256 [0.0017, 0.0495]	0.036	0.072
F	DA-P (proposed)	0.0256 [-0.0511, 0.1023]	0.418	0.837
Random forest vs linear				
R	Paired t (folds)	-0.0298 [-0.0324, -0.0271]	<0.001	<0.001
R	i.i.d. bootstrap	-0.0298 [-0.0327, -0.0270]	0.001	0.003
R	Corrected resampled t	-0.0302 [-0.0338, -0.0265]	<0.001	<0.001
B	Paired t (folds)	-0.0001 [-0.0130, 0.0128]	0.978	1.000
B	i.i.d. bootstrap	-0.0005 [-0.0037, 0.0030]	0.763	0.763
B	Block bootstrap	-0.0005 [-0.0064, 0.0057]	0.875	0.932
B	Mixed model	-0.0005 [-0.0068, 0.0059]	0.878	0.975
B	DM–HAC (time only)	-0.0005 [-0.0052, 0.0042]	0.842	0.842
B	Hering–Genton (space–time)	-0.0005 [-0.0109, 0.0099]	0.928	1.000
B	DA-K (kernel variant)	-0.0005 [-0.0155, 0.0146]	0.941	1.000
B	DA-P (proposed)	-0.0005 [-0.0109, 0.0099]	0.928	1.000
F	Paired t (folds)	-0.0065 [-0.0580, 0.0450]	0.717	0.717
F	Block bootstrap	-0.0064 [-0.0243, 0.0094]	0.438	0.438
F	Hering–Genton (space–time)	-0.0064 [-0.0178, 0.0050]	0.271	0.271
F	DA-P (proposed)	-0.0064 [-0.0777, 0.0649]	0.803	0.837

Table A2 Case study: Melbourne pedestrians (squared error of $\log(1 + \text{count})$).

CV	Procedure	$\hat{\Delta}$ [95% CI]	p	p_{Holm}
XGBoost vs random forest				
R	Paired t (folds)	0.0003 [-0.0024, 0.0030]	0.795	0.795
R	i.i.d. bootstrap	0.0003 [-0.0015, 0.0019]	0.757	0.757
R	Corrected resampled t	0.0003 [-0.0020, 0.0027]	0.782	0.782
B	Paired t (folds)	-0.0025 [-0.0047, -0.0003]	0.033	0.033
B	i.i.d. bootstrap	-0.0025 [-0.0043, -0.0007]	0.011	0.011
B	Block bootstrap	-0.0025 [-0.0090, 0.0027]	0.402	0.402
B	Mixed model	-0.0025 [-0.0086, 0.0036]	0.415	0.415
B	DM–HAC (time only)	-0.0025 [-0.0060, 0.0010]	0.156	0.156
B	Hering–Genton (space–time)	-0.0025 [-0.0093, 0.0043]	0.473	0.473
B	DA-K (kernel variant)	-0.0025 [-0.0061, 0.0011]	0.172	0.172
B	DA-P (proposed)	-0.0025 [-0.0093, 0.0043]	0.473	0.473
F	Paired t (folds)	-0.0009 [-0.0150, 0.0132]	0.855	0.855
F	Block bootstrap	-0.0003 [-0.0089, 0.0067]	0.932	0.932
F	Hering–Genton (space–time)	-0.0003 [-0.0099, 0.0092]	0.947	0.947
F	DA-P (proposed)	-0.0003 [-0.0099, 0.0092]	0.947	0.947
XGBoost vs linear				
R	Paired t (folds)	-1.89 [-1.91, -1.88]	<0.001	<0.001
R	i.i.d. bootstrap	-1.89 [-1.91, -1.88]	0.001	0.003
R	Corrected resampled t	-1.89 [-1.91, -1.87]	<0.001	<0.001
B	Paired t (folds)	-1.85 [-1.90, -1.80]	<0.001	<0.001
B	i.i.d. bootstrap	-1.85 [-1.87, -1.83]	0.001	0.003
B	Block bootstrap	-1.85 [-1.91, -1.79]	0.001	0.003
B	Mixed model	-1.85 [-1.91, -1.78]	<0.001	<0.001
B	DM–HAC (time only)	-1.85 [-1.89, -1.81]	<0.001	<0.001
B	Hering–Genton (space–time)	-1.85 [-1.91, -1.79]	<0.001	<0.001
B	DA-K (kernel variant)	-1.85 [-1.90, -1.80]	<0.001	<0.001
B	DA-P (proposed)	-1.85 [-1.91, -1.79]	<0.001	<0.001
F	Paired t (folds)	-1.83 [-2.15, -1.50]	<0.001	0.001
F	Block bootstrap	-1.83 [-1.91, -1.75]	0.001	0.003
F	Hering–Genton (space–time)	-1.83 [-1.92, -1.74]	<0.001	<0.001
F	DA-P (proposed)	-1.83 [-2.26, -1.41]	<0.001	0.001
Random forest vs linear				
R	Paired t (folds)	-1.89 [-1.91, -1.88]	<0.001	<0.001
R	i.i.d. bootstrap	-1.89 [-1.91, -1.87]	0.001	0.003
R	Corrected resampled t	-1.89 [-1.91, -1.87]	<0.001	<0.001
B	Paired t (folds)	-1.85 [-1.90, -1.79]	<0.001	<0.001
B	i.i.d. bootstrap	-1.85 [-1.87, -1.83]	0.001	0.003
B	Block bootstrap	-1.85 [-1.91, -1.78]	0.001	0.003
B	Mixed model	-1.85 [-1.91, -1.78]	<0.001	<0.001
B	DM–HAC (time only)	-1.85 [-1.88, -1.81]	<0.001	<0.001
B	Hering–Genton (space–time)	-1.85 [-1.91, -1.78]	<0.001	<0.001
B	DA-K (kernel variant)	-1.85 [-1.90, -1.79]	<0.001	<0.001
B	DA-P (proposed)	-1.85 [-1.91, -1.78]	<0.001	<0.001
F	Paired t (folds)	-1.83 [-2.16, -1.49]	<0.001	0.001
F	Block bootstrap	-1.83 [-1.90, -1.75]	0.001	0.003
F	Hering–Genton (space–time)	-1.83 [-1.92, -1.75]	<0.001	<0.001
F	DA-P (proposed)	-1.83 [-2.27, -1.40]	<0.001	0.001

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Table A3 Case study: Las Rosas maize (squared error of yield, (q/ha)^2).

CV	Procedure	$\hat{\Delta}$ [95% CI]	p	p_{Holm}
XGBoost vs random forest				
R	Paired t (folds)	-2.82 [-4.64, -0.99]	0.013	0.013
R	i.i.d. bootstrap	-2.82 [-3.67, -1.81]	0.001	0.003
R	Corrected resampled t	-2.36 [-3.53, -1.19]	<0.001	<0.001
B	Paired t (folds)	-20.9 [-70.4, 28.7]	0.308	0.308
B	i.i.d. bootstrap	-21.1 [-24.4, -18.1]	0.001	0.003
B	Block bootstrap	-21.1 [-47.0, -3.3]	0.043	0.043
B	Mixed model	-21.1 [-44.0, 1.8]	0.069	0.069
B	Hering–Genton (space–time)	-21.1 [-84.2, 41.9]	0.511	0.511
B	DA-K (kernel variant)	-21.1 [-88.3, 46.1]	0.449	0.449
B	DA-P (proposed)	-21.1 [-80.7, 38.4]	0.459	0.459
XGBoost vs linear				
R	Paired t (folds)	-131.9 [-141.2, -122.7]	<0.001	<0.001
R	i.i.d. bootstrap	-131.9 [-138.1, -126.0]	0.001	0.003
R	Corrected resampled t	-131.7 [-137.5, -125.9]	<0.001	<0.001
B	Paired t (folds)	-160.0 [-193.0, -127.1]	<0.001	<0.001
B	i.i.d. bootstrap	-160.0 [-167.7, -152.1]	0.001	0.003
B	Block bootstrap	-160.0 [-210.4, -112.3]	0.001	0.003
B	Mixed model	-160.0 [-210.0, -109.9]	<0.001	<0.001
B	Hering–Genton (space–time)	-160.0 [-263.1, -56.9]	0.002	0.007
B	DA-K (kernel variant)	-160.0 [-210.3, -109.6]	<0.001	<0.001
B	DA-P (proposed)	-160.0 [-277.1, -42.8]	0.011	0.032
Random forest vs linear				
R	Paired t (folds)	-129.1 [-138.8, -119.3]	<0.001	<0.001
R	i.i.d. bootstrap	-129.1 [-134.9, -123.1]	0.001	0.003
R	Corrected resampled t	-129.4 [-135.1, -123.6]	<0.001	<0.001
B	Paired t (folds)	-133.6 [-229.6, -37.7]	0.018	0.036
B	i.i.d. bootstrap	-133.2 [-142.5, -124.0]	0.001	0.003
B	Block bootstrap	-133.2 [-195.2, -67.1]	0.001	0.003
B	Mixed model	-133.2 [-202.6, -63.8]	<0.001	<0.001
B	Hering–Genton (space–time)	-133.2 [-264.3, -2.0]	0.047	0.093
B	DA-K (kernel variant)	-133.2 [-244.1, -22.3]	0.024	0.047
B	DA-P (proposed)	-133.2 [-261.5, -4.8]	0.043	0.086